

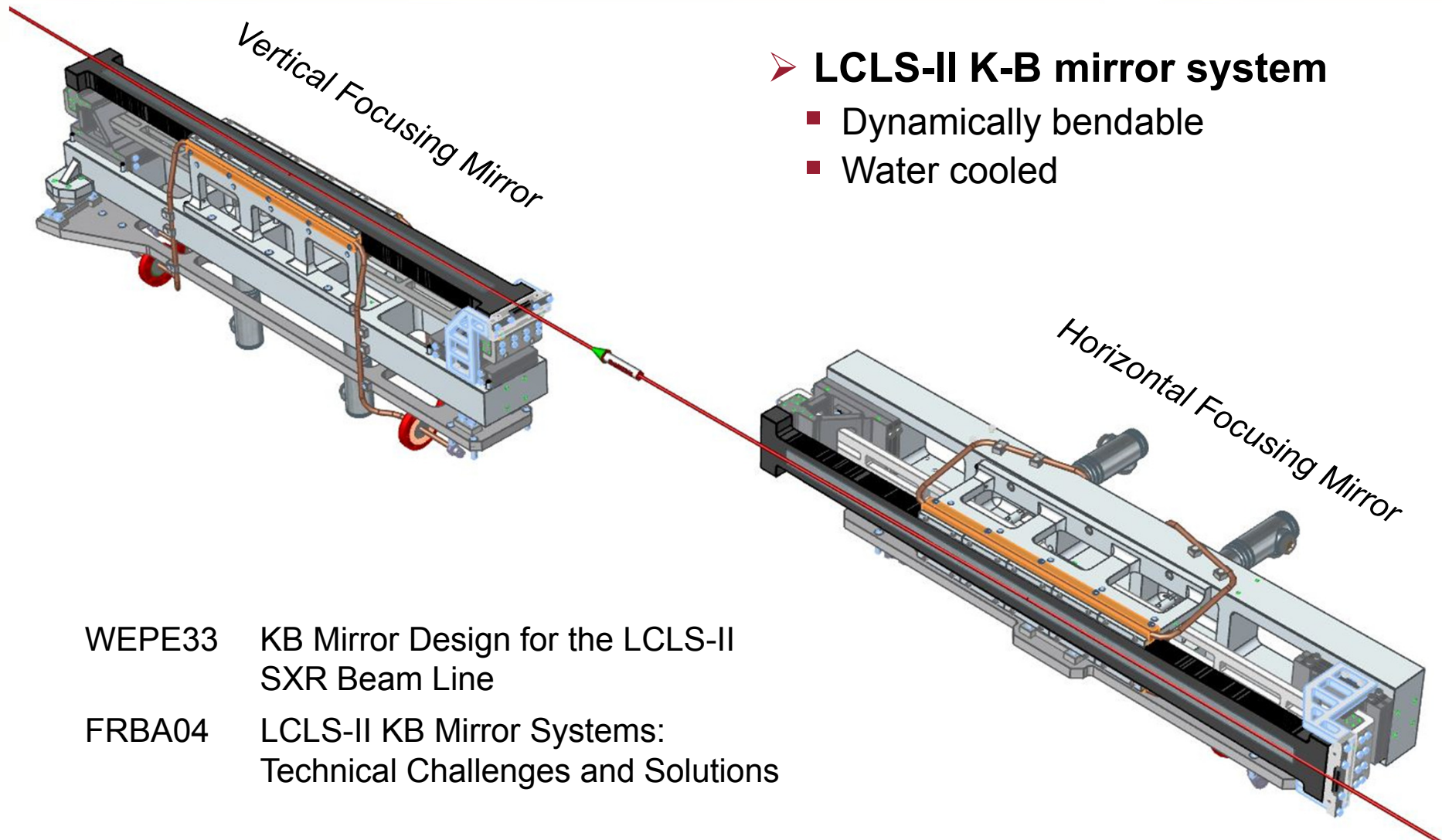
Minimization of mechanical constraint effects of Eutectic Galn as thermal interface

L. Zhang, D. Cocco, N. Kelez, J. James, D.S. Morton

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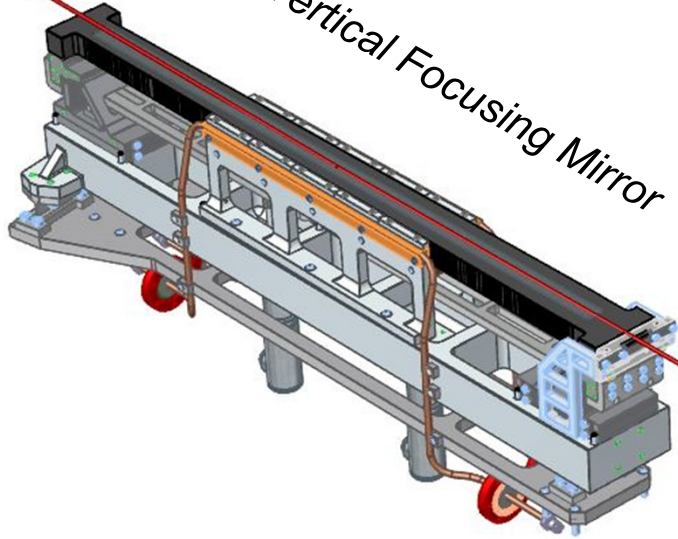
zhanglin@slac.stanford.edu



WEPE33 KB Mirror Design for the LCLS-II
SXR Beam Line

FRBA04 LCLS-II KB Mirror Systems:
Technical Challenges and Solutions

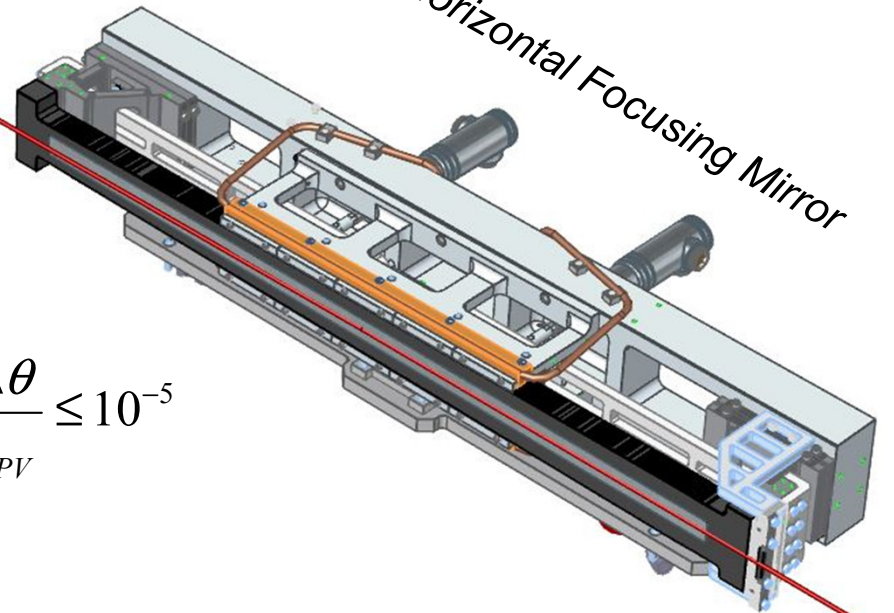
Vertical Focusing Mirror



➤ LCLS-II K-B mirror system

- Dynamically bendable
- Water cooled

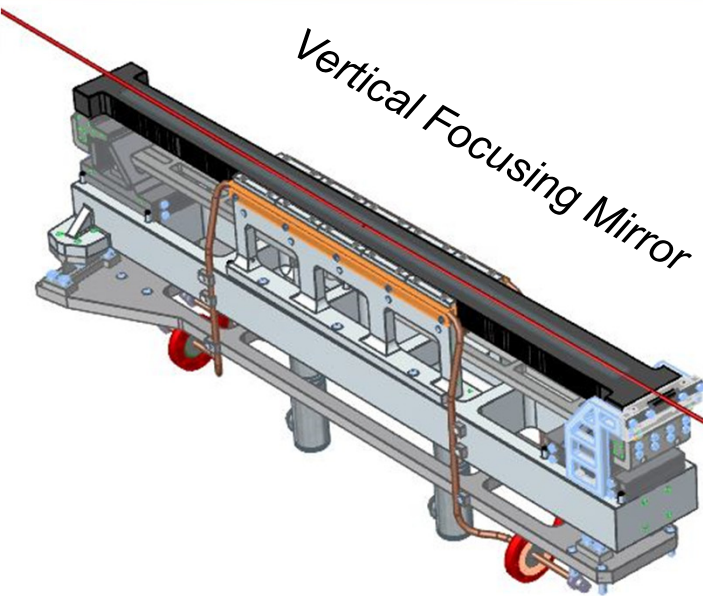
Horizontal Focusing Mirror



➤ Requirements

- Accurate bent mirror shape
- Effective cooling

$$\frac{\Delta\theta}{\theta_{PV}} \leq 10^{-5}$$



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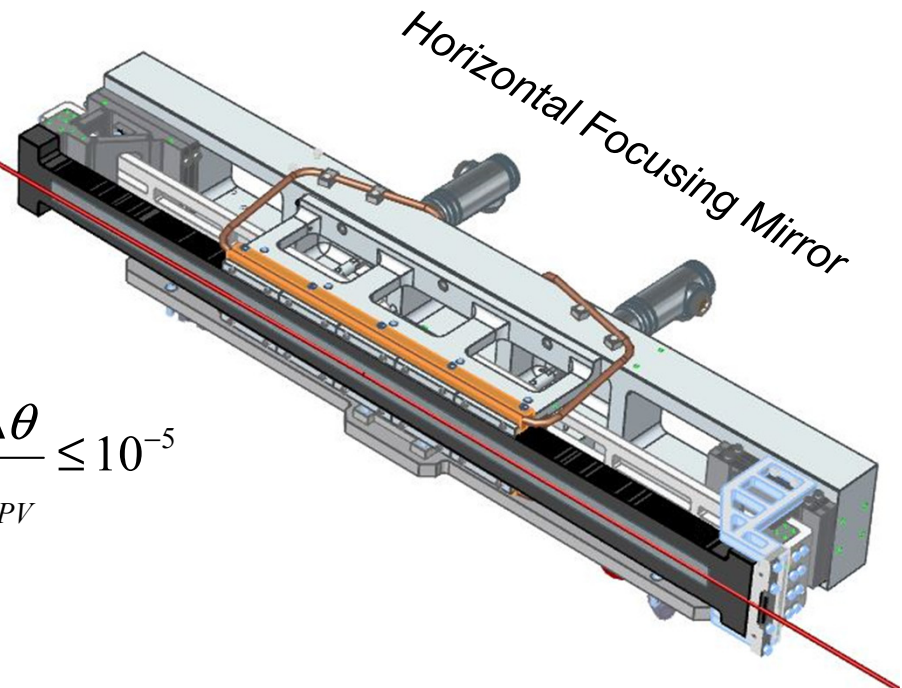
➤ Requirements

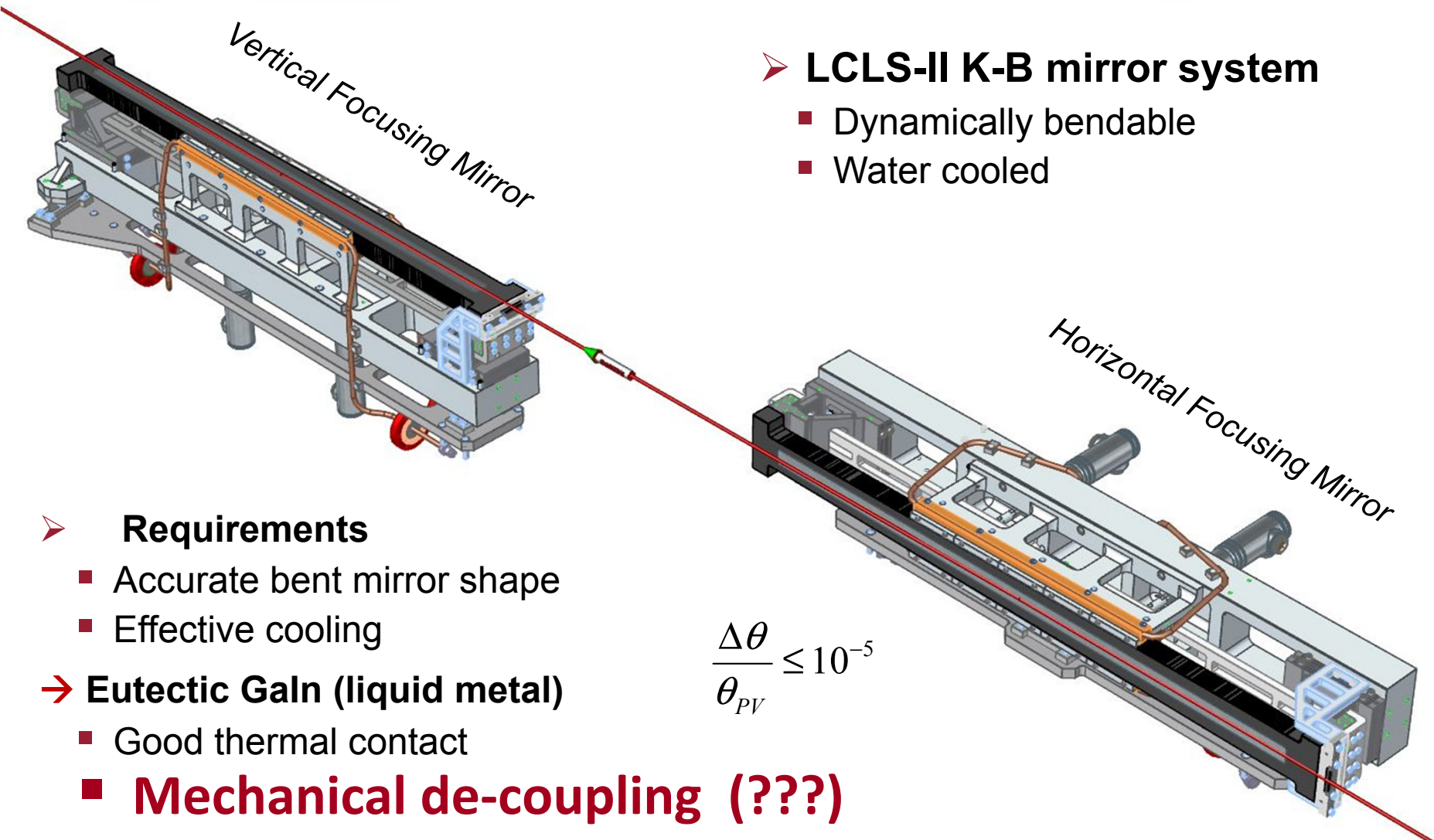
- Accurate bent mirror shape
- Effective cooling

➔ Eutectic Galn (liquid metal)

- Good thermal contact
- Mechanical de-coupling

$$\frac{\Delta\theta}{\theta_{PV}} \leq 10^{-5}$$





➤ LCLS-II K-B mirror system

- Dynamically bendable
- Water cooled

➤ Requirements

- Accurate bent mirror shape
- Effective cooling

➔ Eutectic Galn (liquid metal)

- Good thermal contact
- **Mechanical de-coupling (???)**

$$\frac{\Delta\theta}{\theta_{PV}} \leq 10^{-5}$$

Eutectic Gallium-Indium (eGaln)

- **eGaln: 75.5% Gallium (29.76°C)
24.5% Indium (156.6°C)** $T_{\text{melting}}=15.7^{\circ}\text{C}$



- **Thermal interface**
 - Thermal conductance $> 10^5 \text{ W/m}^2\cdot\text{K}$, ~ 10 times better than Indium foil
 - Interface or trough / bath for indirect cooling of the X-ray mirrors

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- **Mechanical properties**

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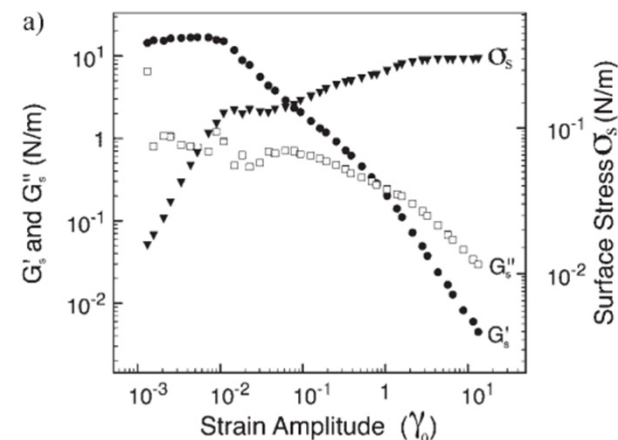


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- **Mechanical properties**

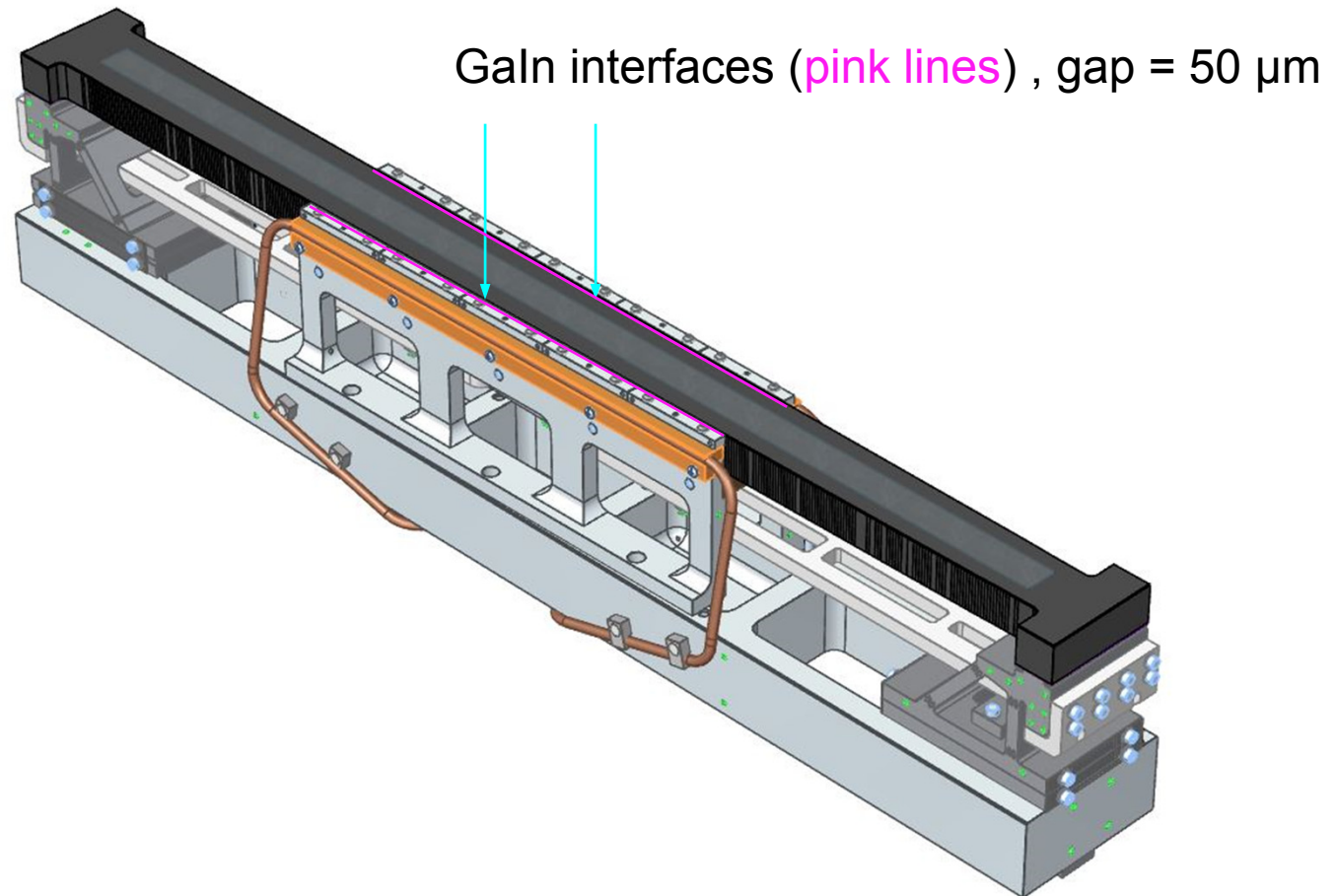
- Bulk viscosity $\sim 2 \text{ mPa}\cdot\text{s}$ ($2\mu_{\text{water}}$)
- Modulus ??
- Thin skin (oxides of Ga)



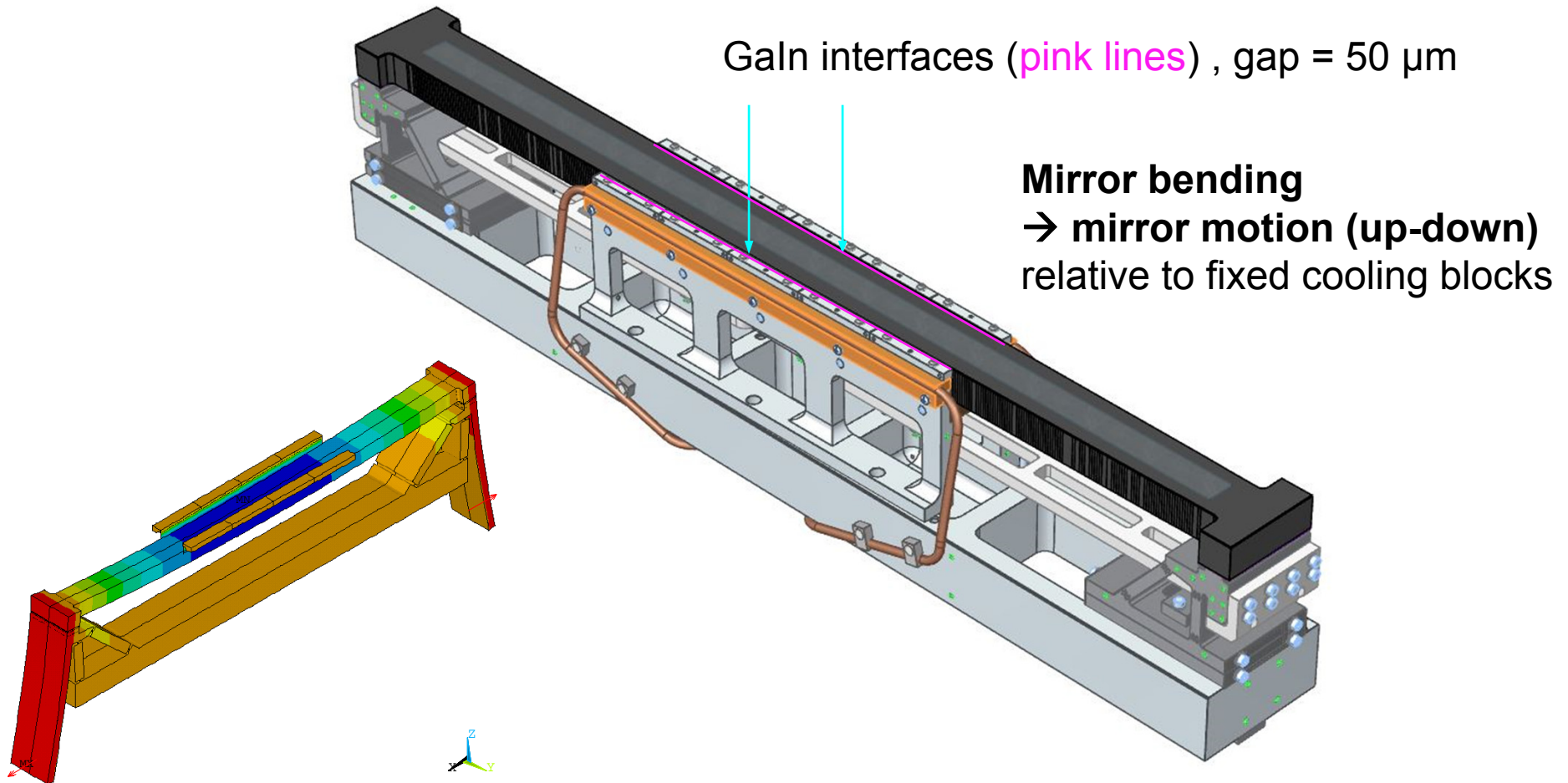
Dickey et al., Adv.Funct.Mater.2008,18,1097–1104

Dickey, ACSAppl.Mater.Interfaces2014,6,18369–18379

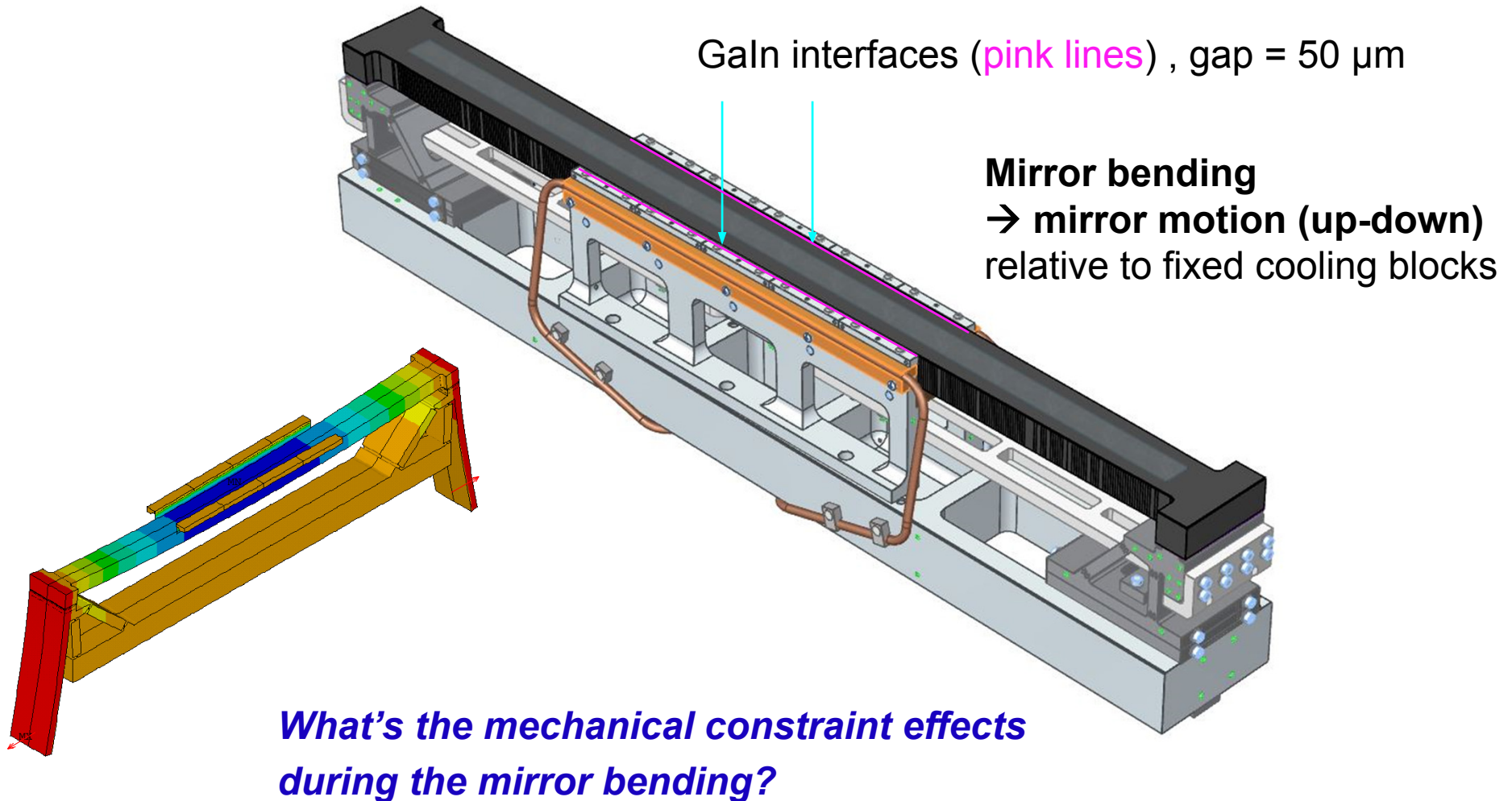
Eutectic Galn as thermal contact interface



Eutectic Galn as thermal contact interface



Eutectic Galn as thermal contact interface

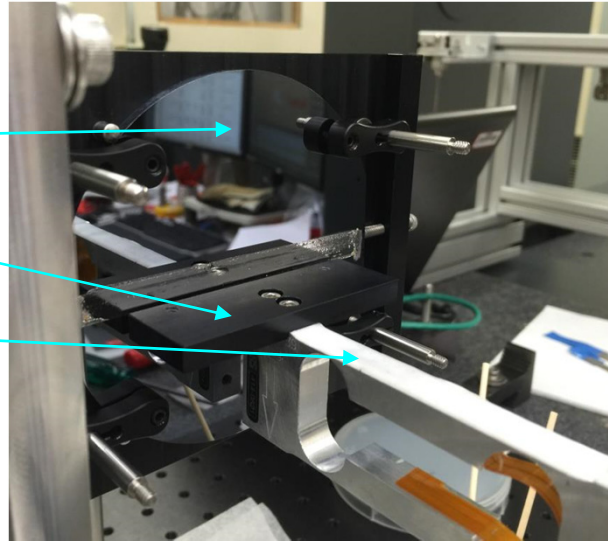


Galn test Setup

- Si-wafer (D=4", $t = 4$ mm, both side polished)
- Galn gap: (51, 102, 152, 203, 254 μm)

*Silicon wafer
moving part*

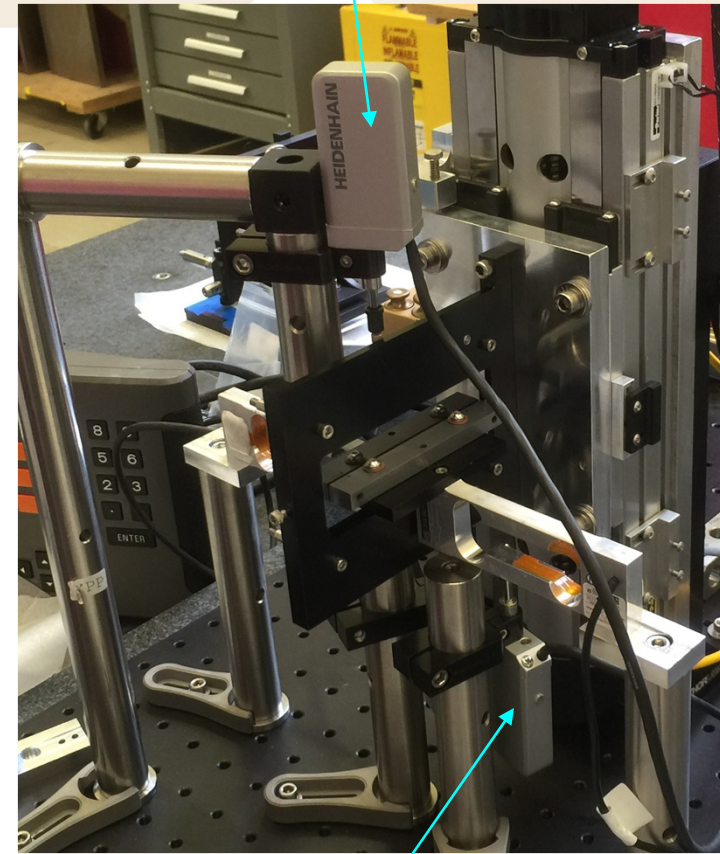
*Silicon pad
fixed on a
force sensor*



➤ Measurements

- Displacement of Si-wafer d_{wafer}
- Forces of Si-pads on
Left side and Right side: F_L, F_R
- Relative displacement between wafer and pads: $d_{\text{pad}} = d_{\text{wafer}} - (F - F_0) / k_{FG}$
- Force gauge stiffness: $k_{FG} = 10.57 \text{ N/mm}$

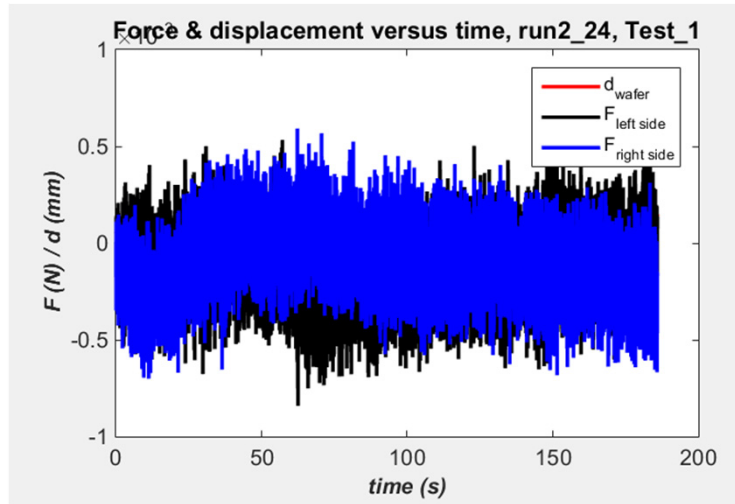
Displ. Sensor A



Displ. Sensor B

Galn test Setup – measurement accuracy

- Force & displacement measured when there is no motion

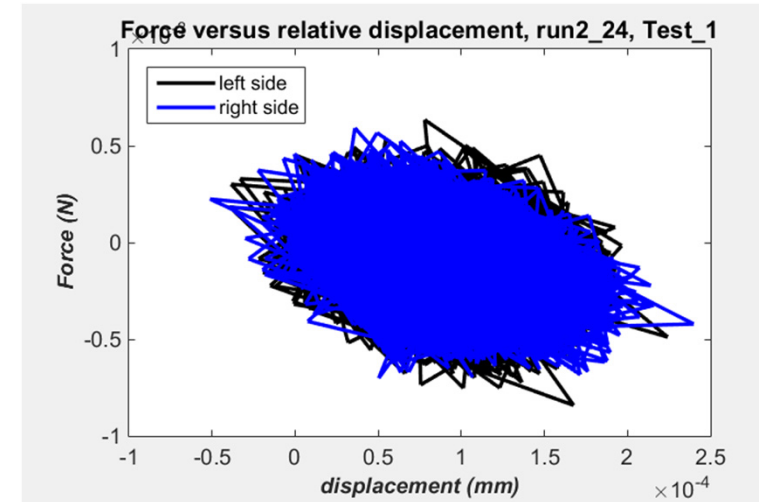


gap = $100\ \mu m$

$\epsilon_d = \pm 0.15\ \mu m$

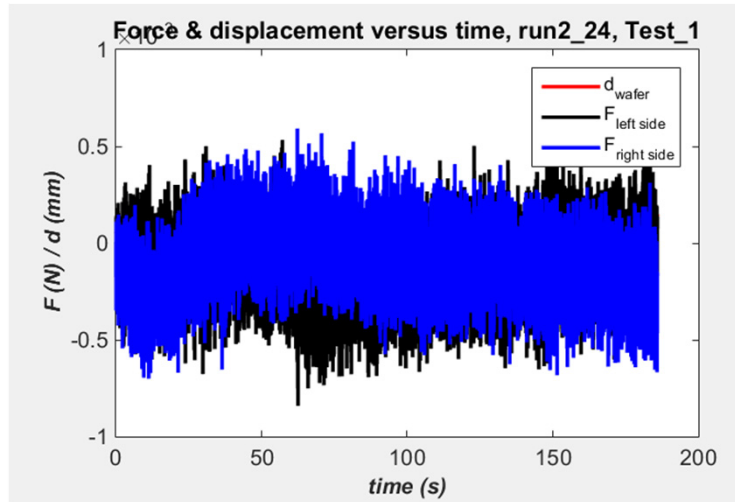
$\epsilon_F = \pm 0.7\ mN$

Sensors accuracy:
 $0.2\ \mu m, 0.2\ mN$



Gain test Setup – measurement accuracy

- Force & displacement measured when there is no motion

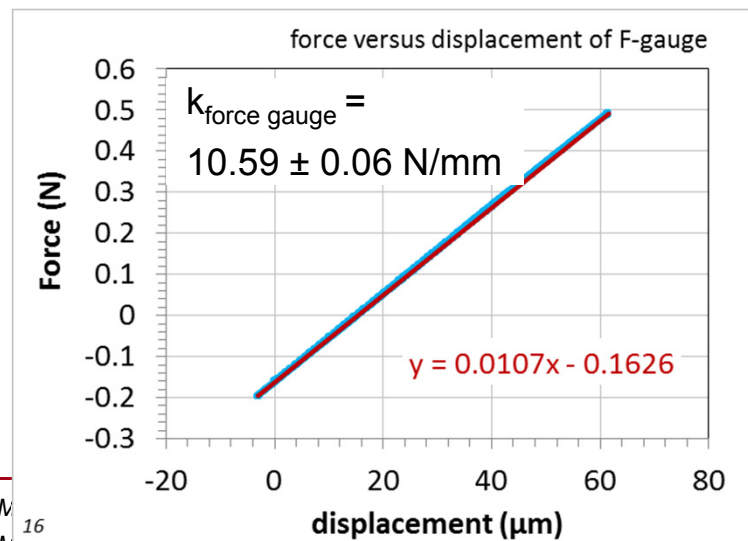
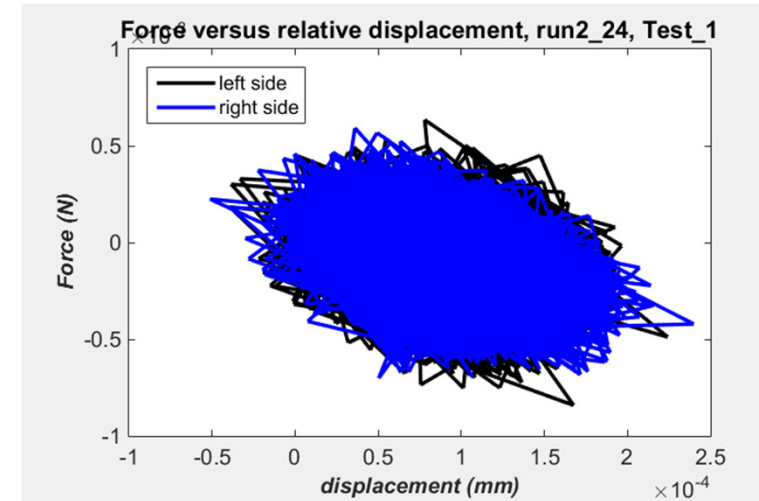


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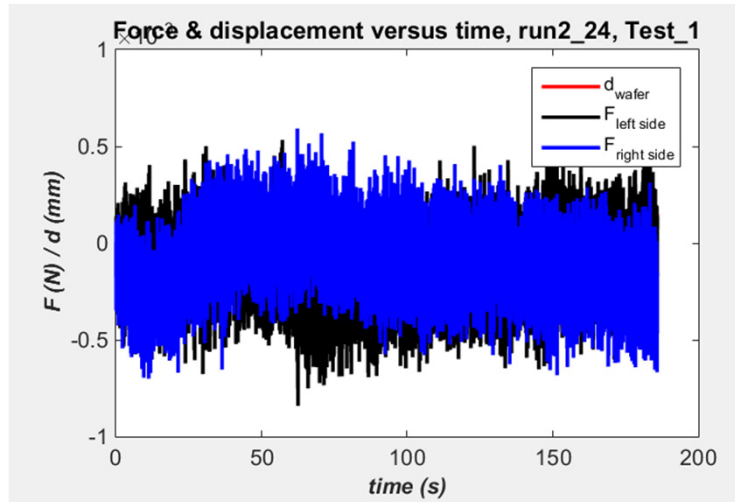
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l interface

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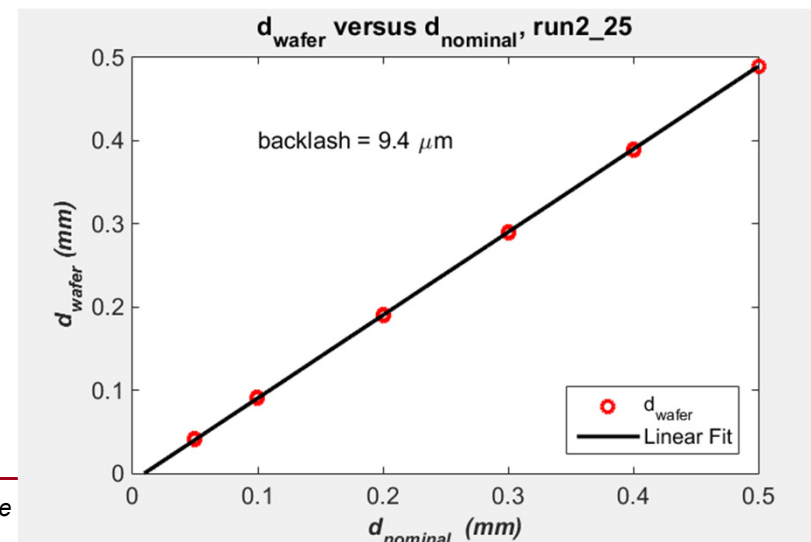
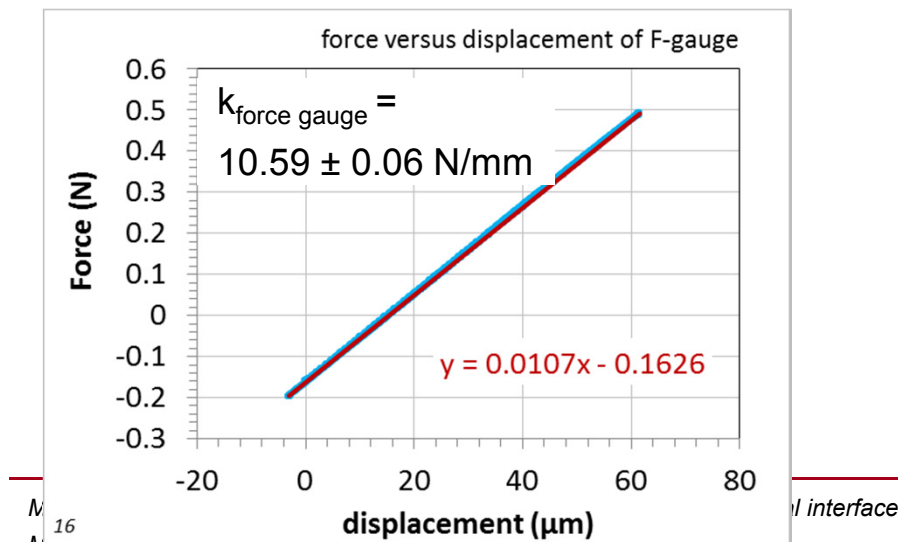
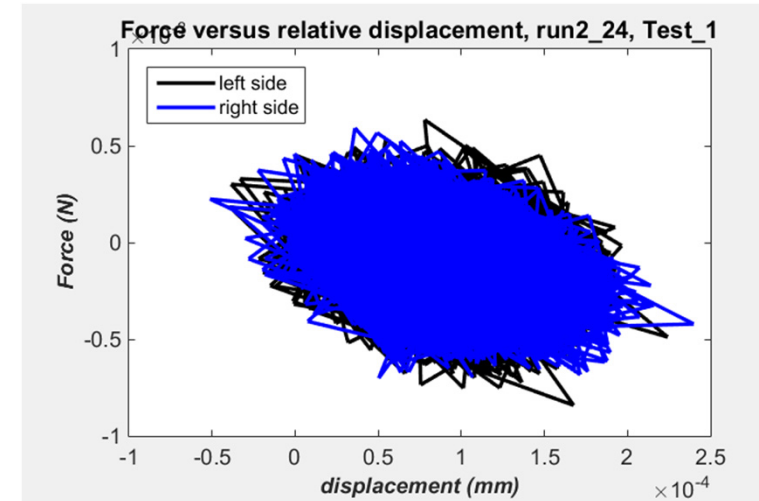


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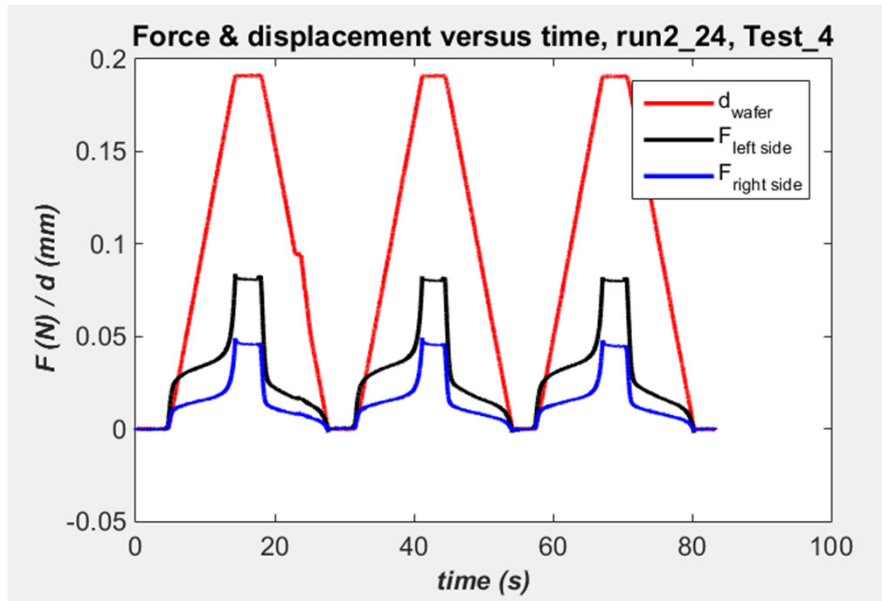
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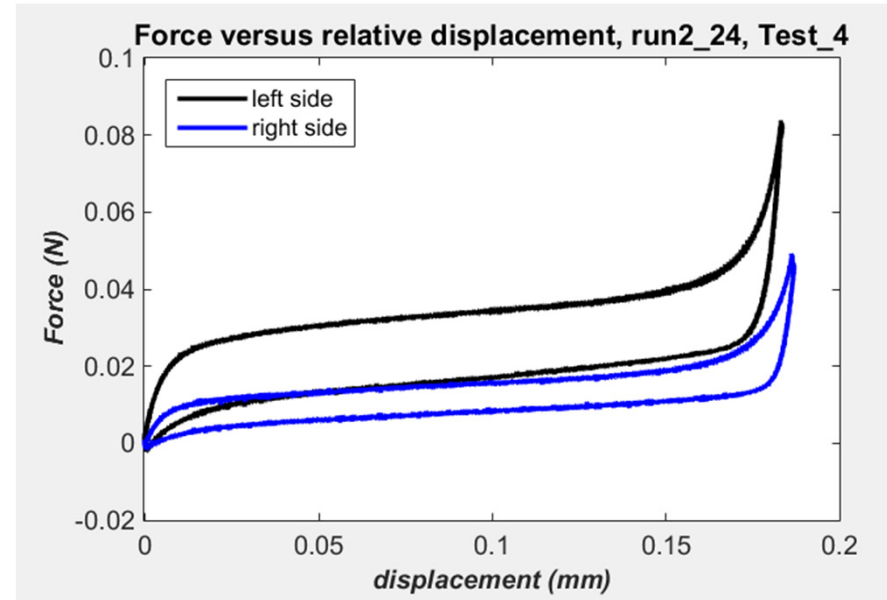
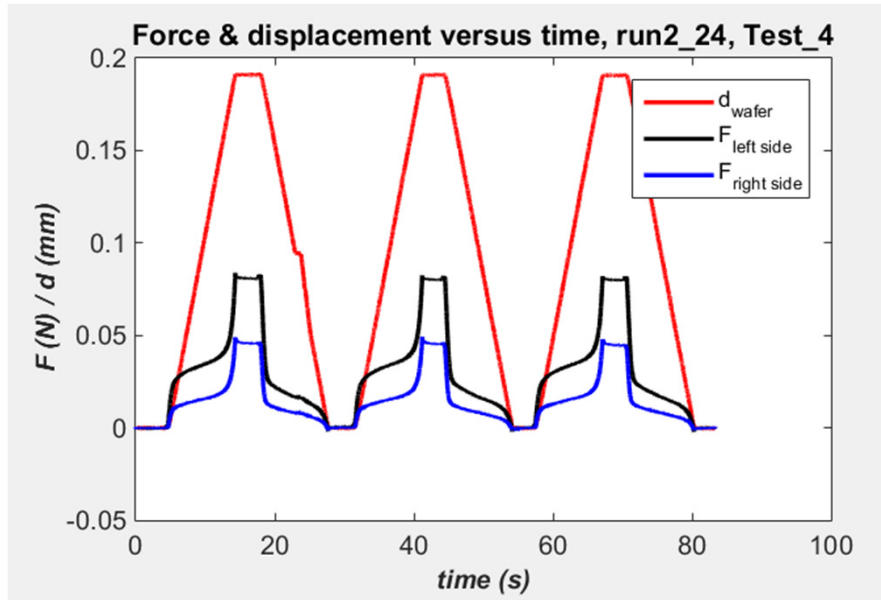
Galn Test results – Repeatability



- Galn gap = $100\ \mu\text{m}$
- Si-wafer displacement $d_{\text{wafer}} = 200\ \mu\text{m}$, at speed $20\ \mu\text{m/s}$
- 3 – cycles measurements → **Good repeatability**

Galn Test results – Repeatability

Share stress $\sim F/A_{\text{contact}}$

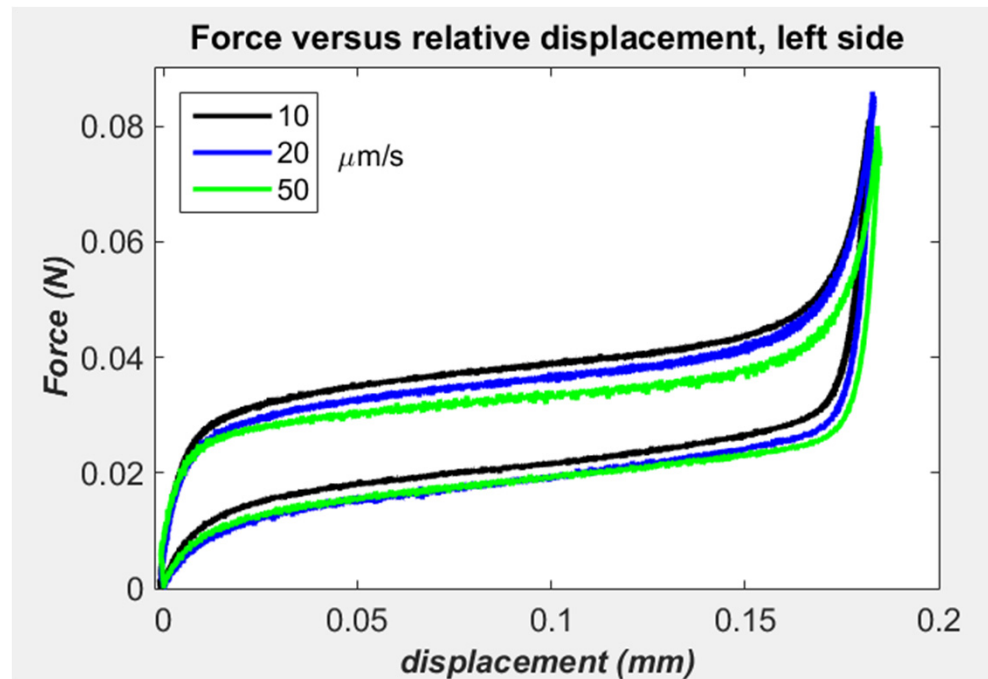


Share strain $\sim d/\text{gap}_{\text{Galn}}$

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- 3 – cycles measurements → **Good repeatability**

Galn Test results – at different speed of displacement

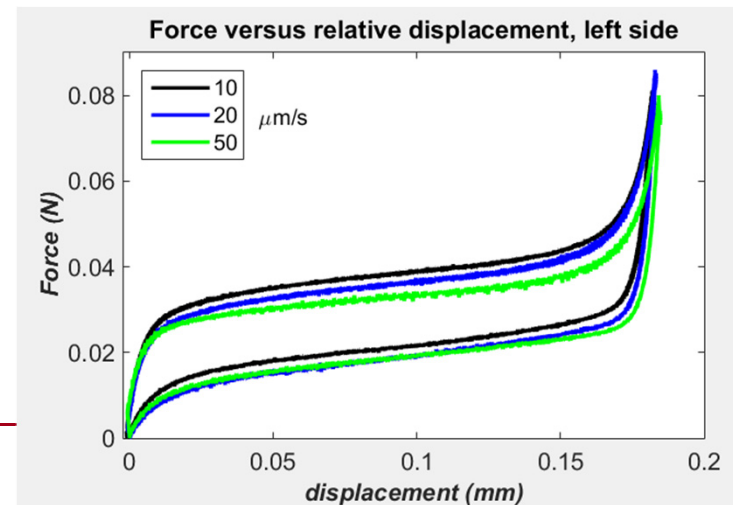
- Si-wafer displacement $d_{\text{wafer}} = 200 \mu\text{m}$, at speed 10, 20, 50 $\mu\text{m/s}$
- 3 cycles measurements per speed → Repeatable



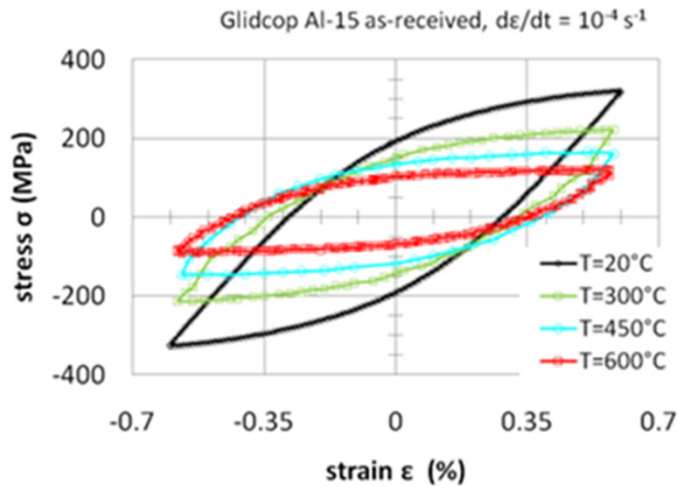
- Speed 20 $\mu\text{m/s}$ is mostly used in other measurements

Which type of material model ?

Shear strain: 100% ~ 400%



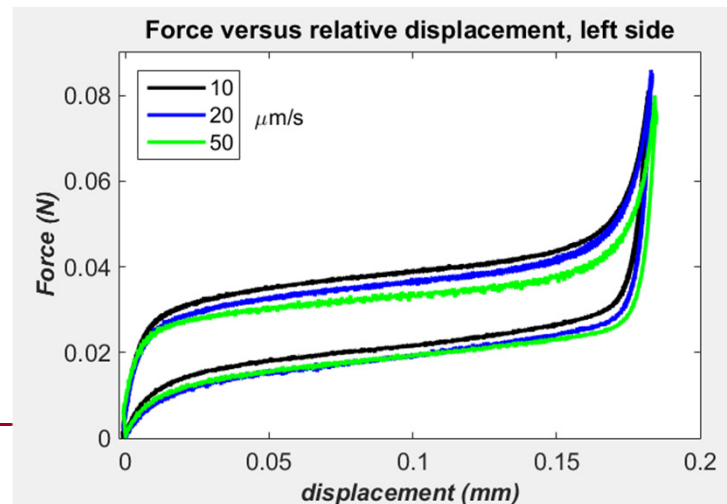
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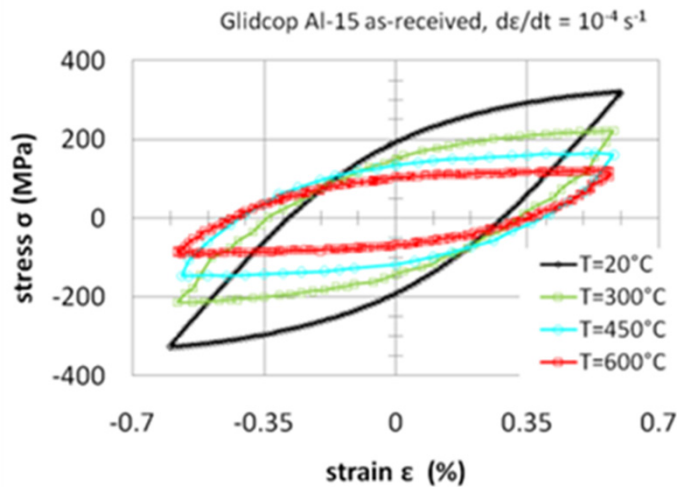
ElastoViscoPlasticity

Glidcop Al-15 at
different temperature

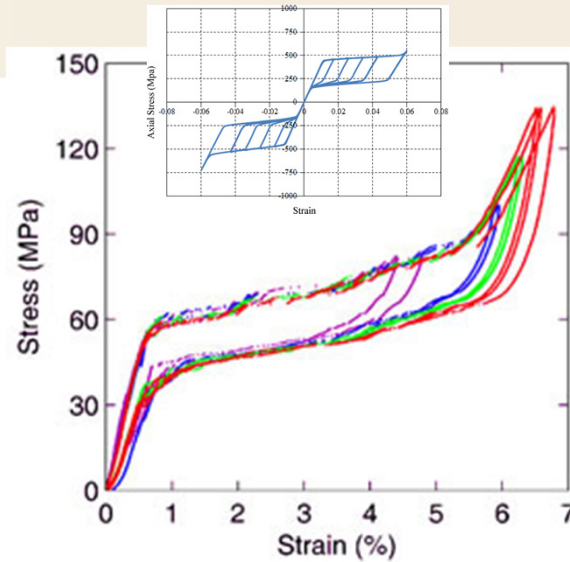
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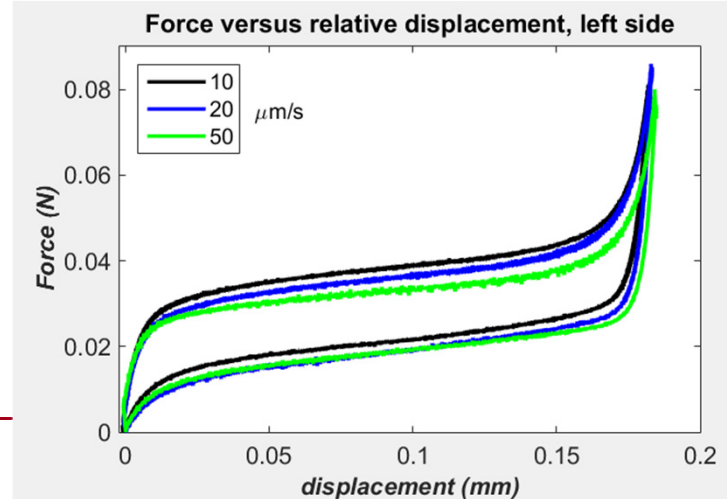


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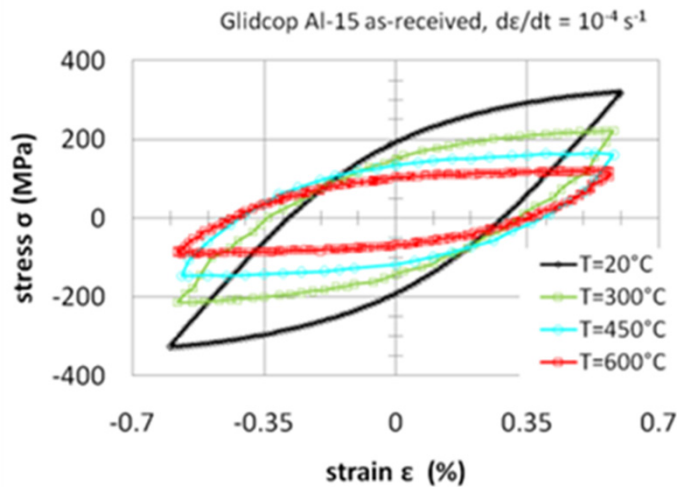


SuperElasticity
NiTi shape-memory
alloys (SMAs)

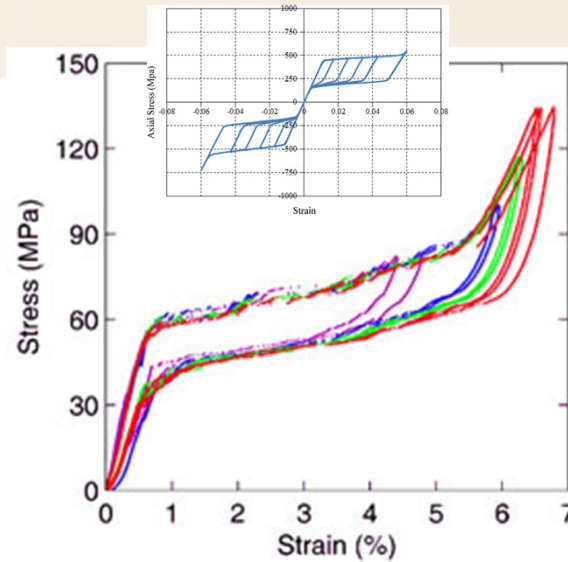
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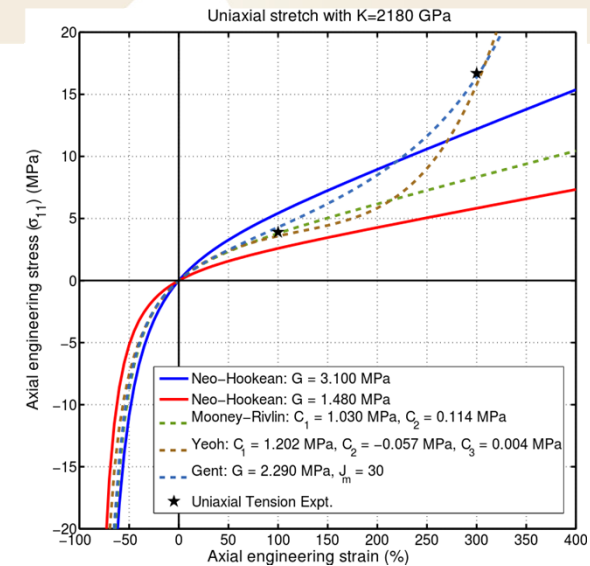
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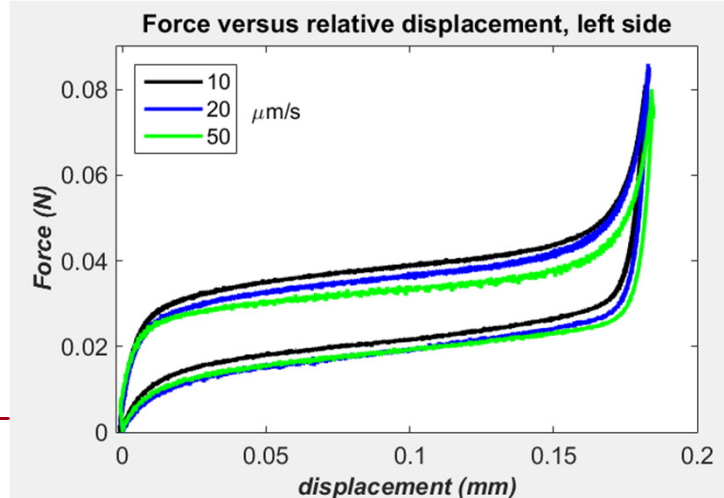


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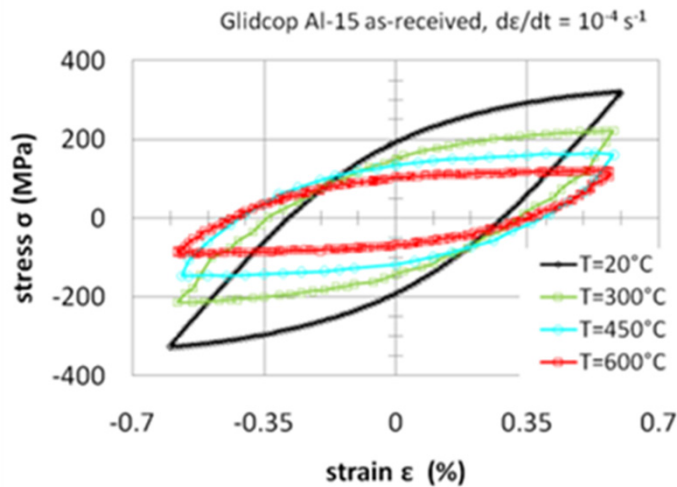


HyperElasticity
 rubber

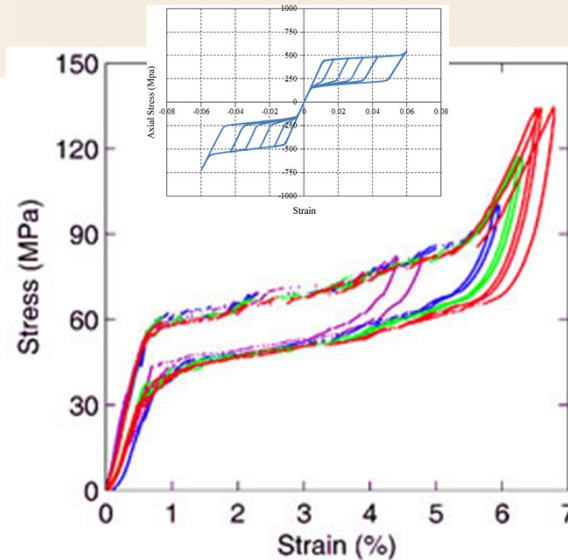
Shear strain: 100% ~ 400%



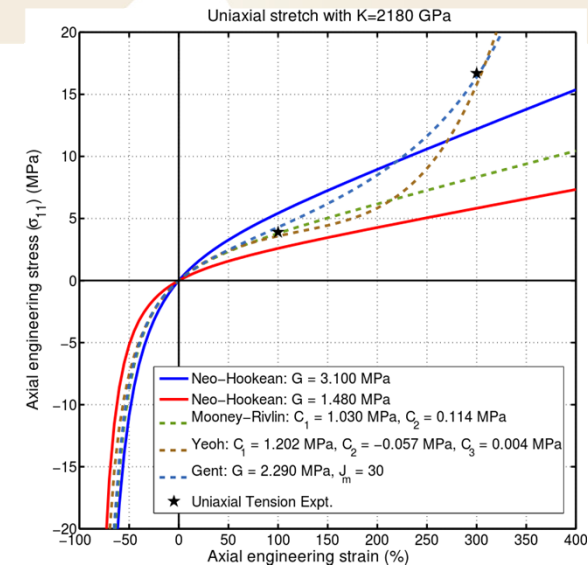
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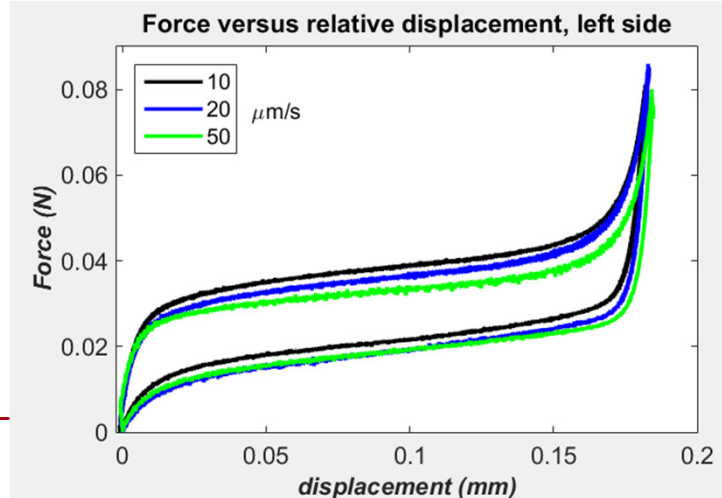


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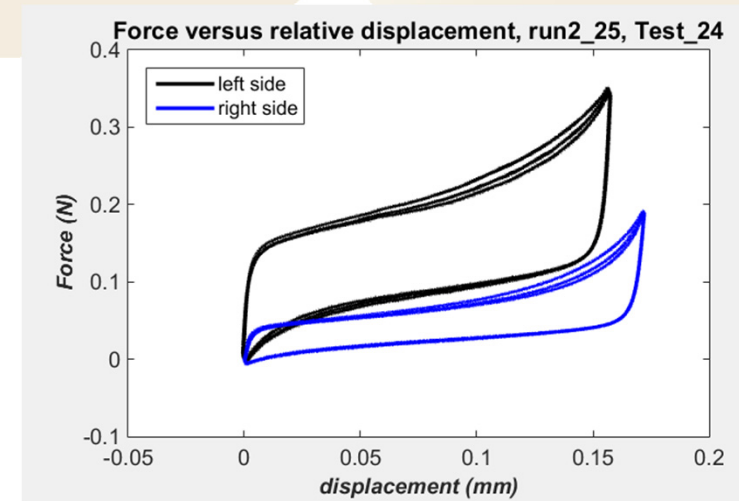
HyperElasticity
 rubber

Shear strain: 100% ~ 400%
 → Full range FEA impossible (non
 convergent) with ANSYS



➤ Mechanical constraint effects

- eGaln interface induced
 - Force $F_{\text{Galn}} \sim 0.35 \text{ N/pad}$
 - $M_{\text{Galn}} \sim F_{\text{Galn}} (2 \cdot 0.35 + 2 \cdot 0.45) = \mathbf{0.56 \text{ Nm}}$
- Bending moments (30-mm thick mirror)
 - $M_{\text{bending}} \sim 0.3 \cdot 90 = \mathbf{27 \text{ Nm}}$
- $M_{\text{Galn}} / M_{\text{bending}} \sim \mathbf{2\%}$ (very significant)



➤ eGaln Viscosity related force

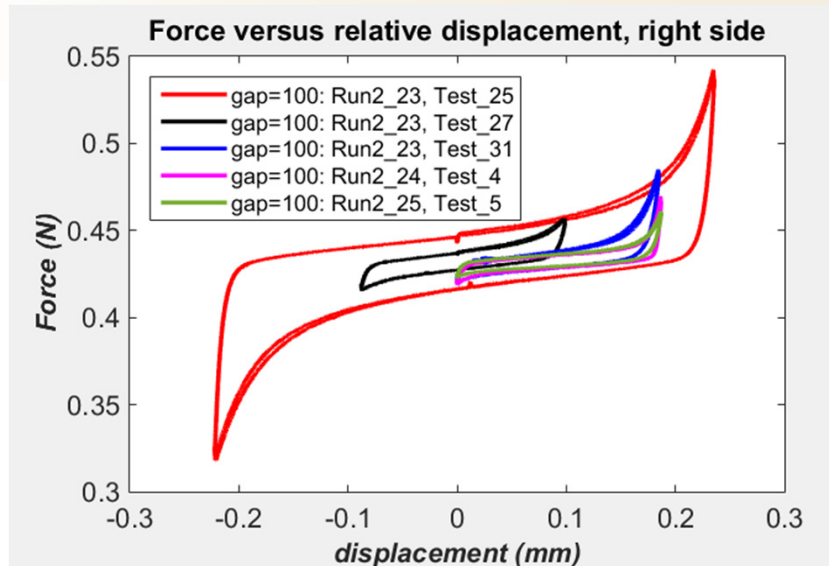
- $F_{\text{viscosity}} \sim 0.4 \text{ } \mu\text{N/pad}$
- $F_{\text{viscosity}} / F_{\text{measured}} \sim 10^{-6}$ (negligible)

$$F_{\text{viscosity}} = A\mu \frac{\partial V}{\partial y}$$

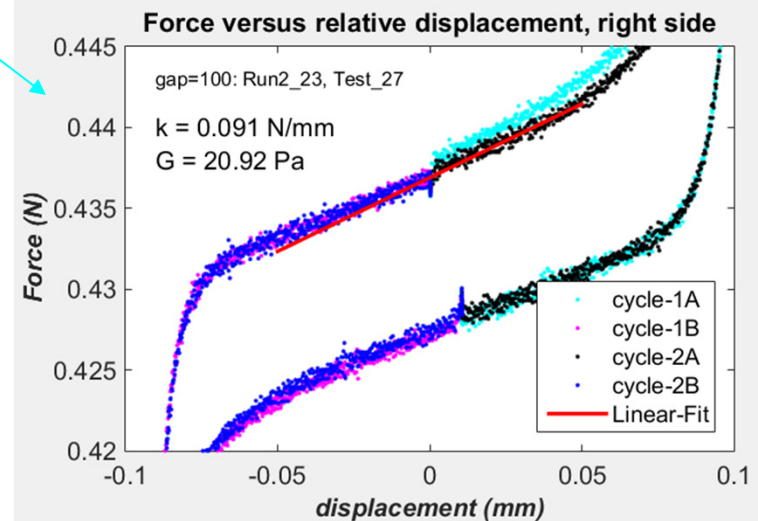
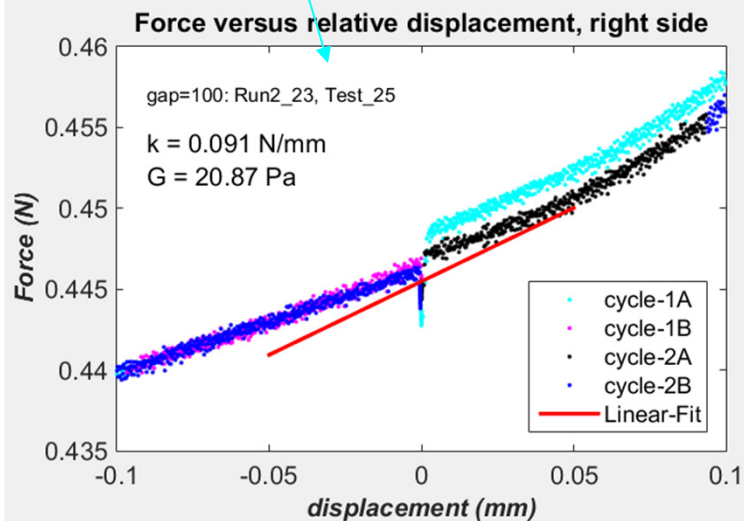
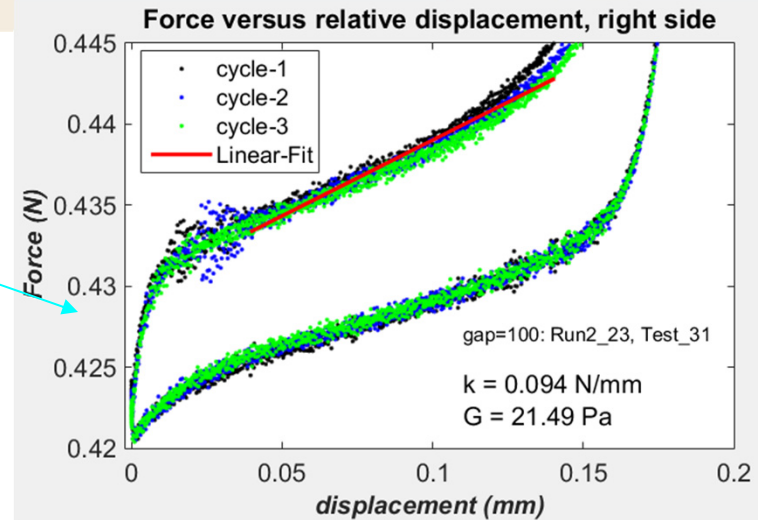
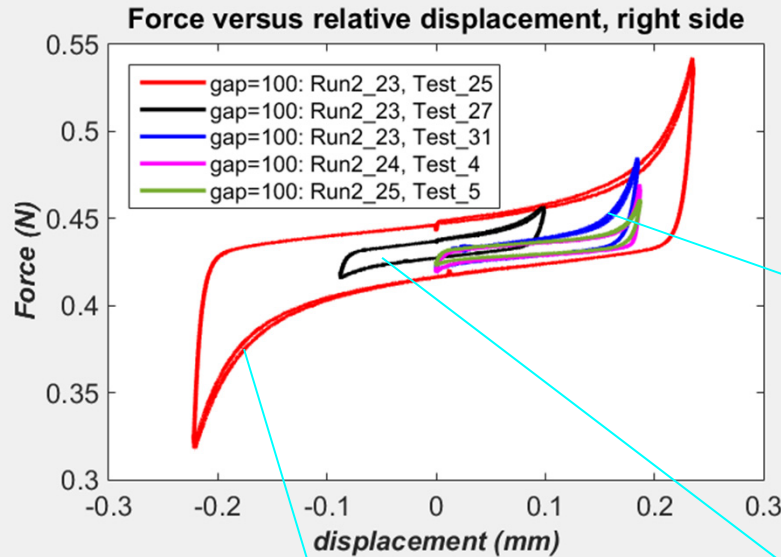
$$g_{\text{Galn}} = 50 \text{ } \mu\text{m}$$

- ## ➤ The mechanical constraint effects of eGaln interface is **not negligible (at this stage)**, but it's **not viscosity related force** (and therefore will not be treated as fluid)

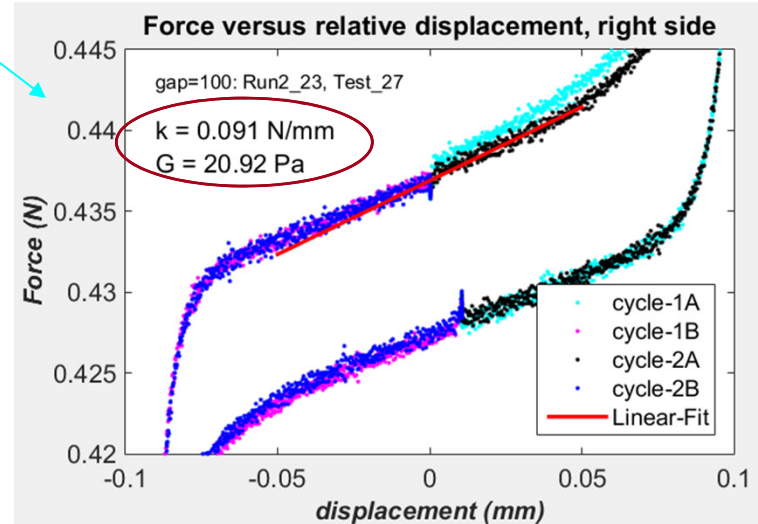
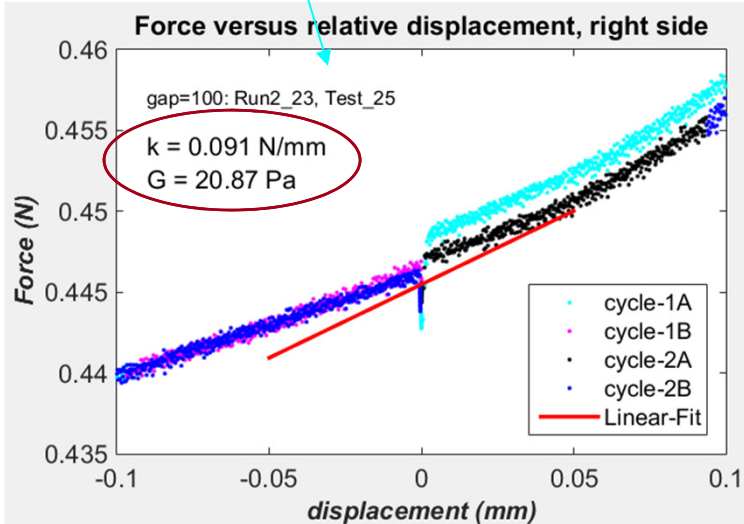
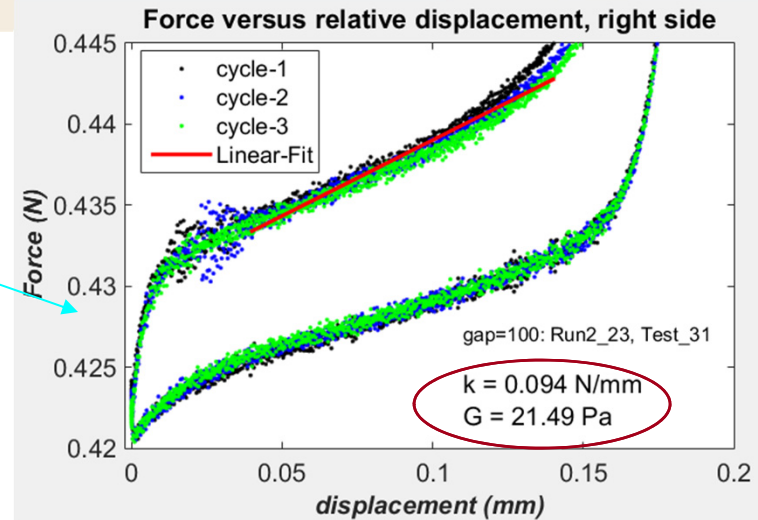
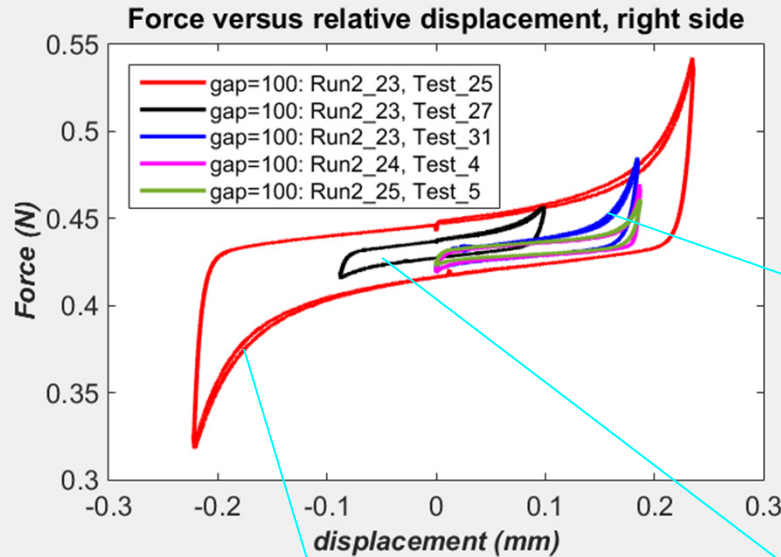
Galn Test results – Data fitting, modulus



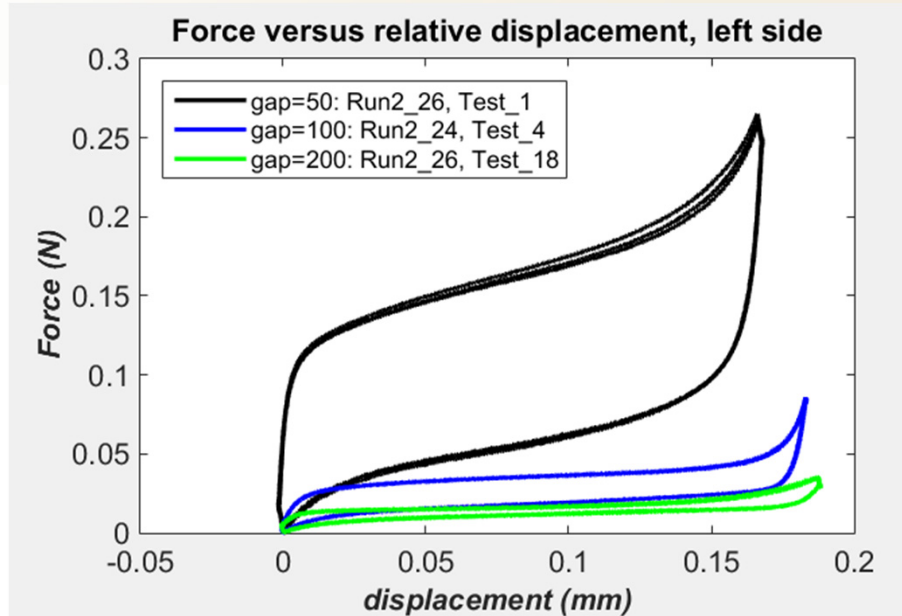
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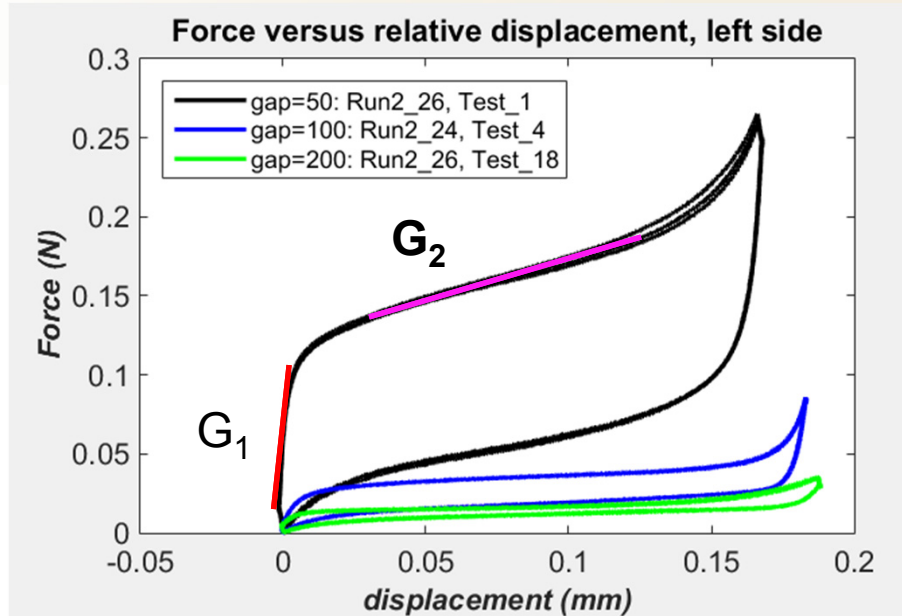
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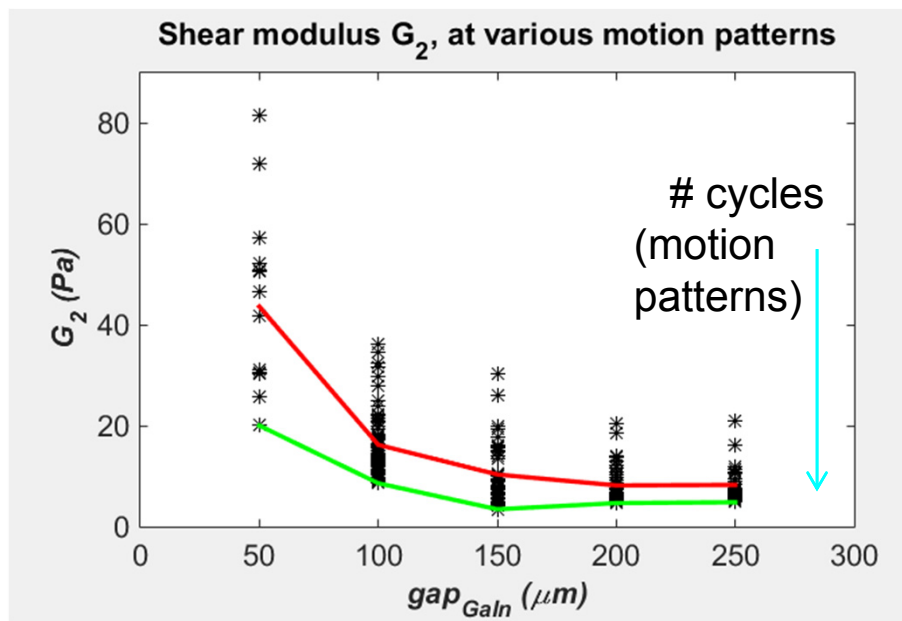
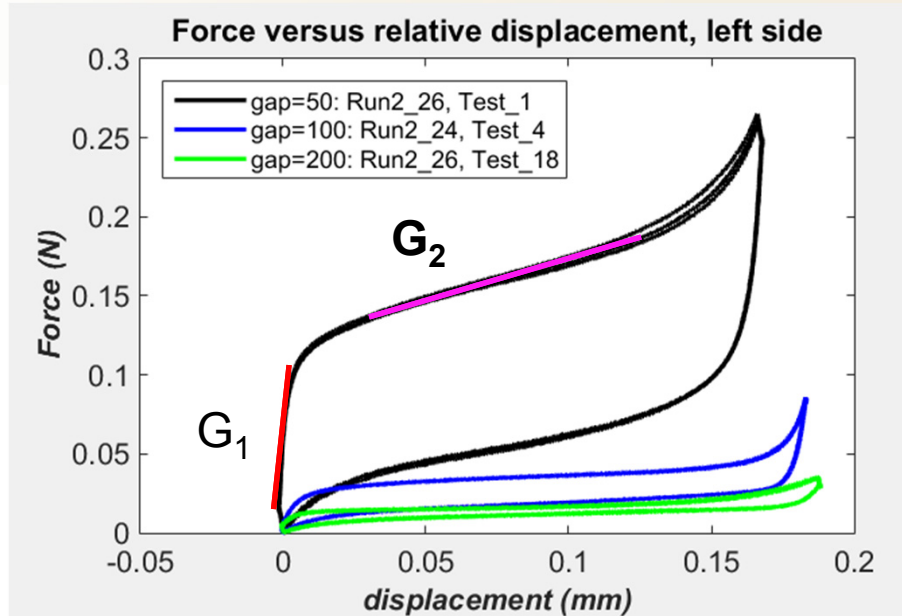
Galn test results – gap dependant; Modulus G_1 and G_2



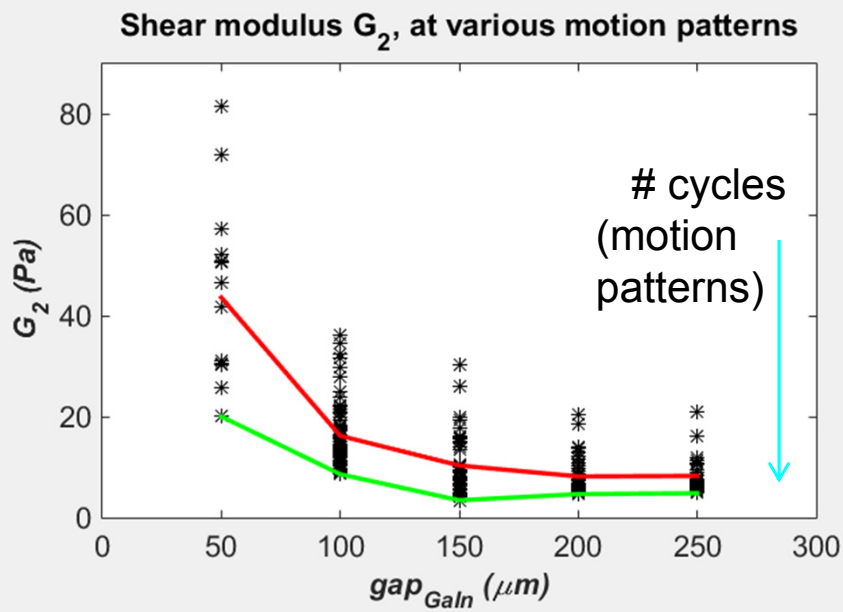
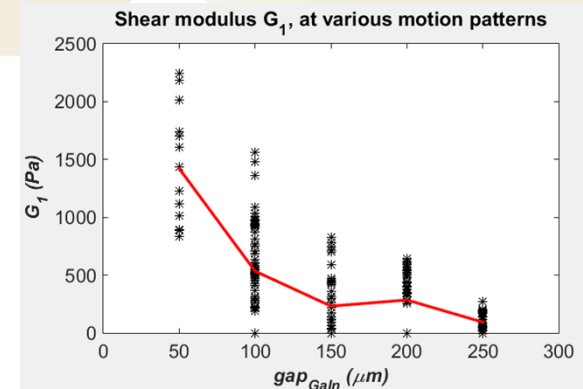
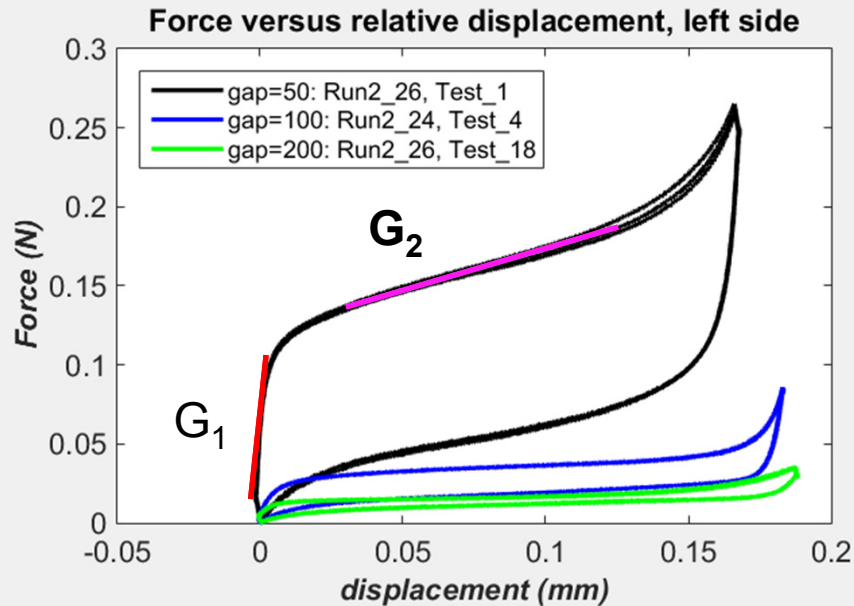
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Gain test results – gap dependant; Modulus G_1 and G_2



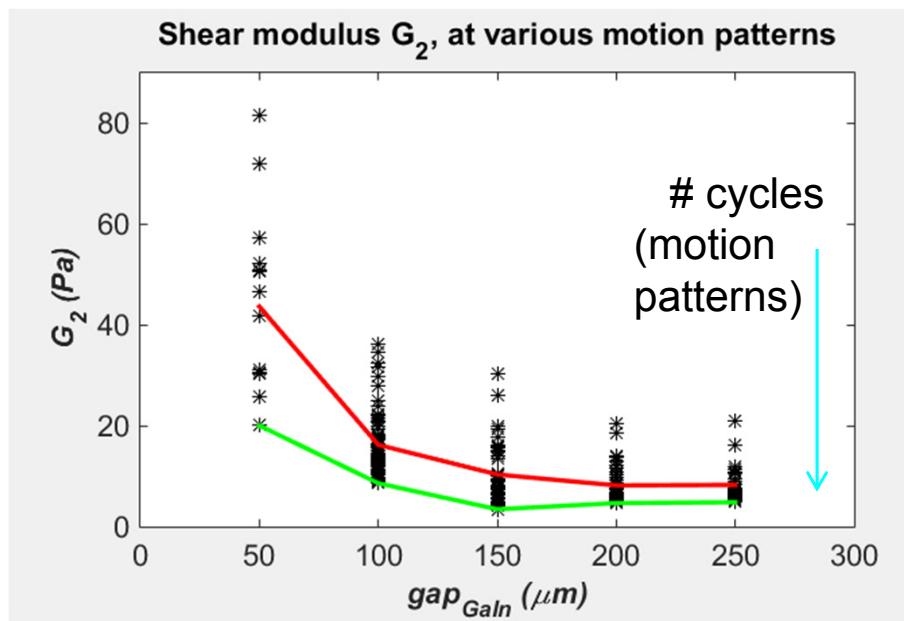
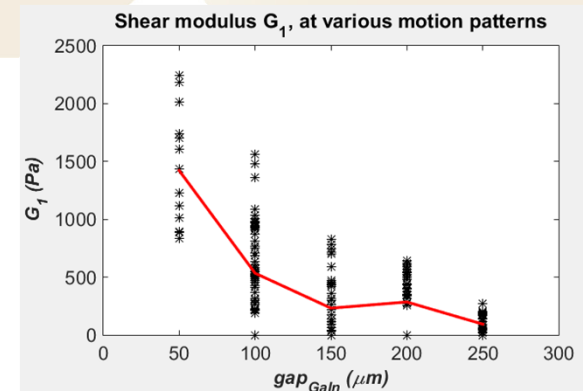
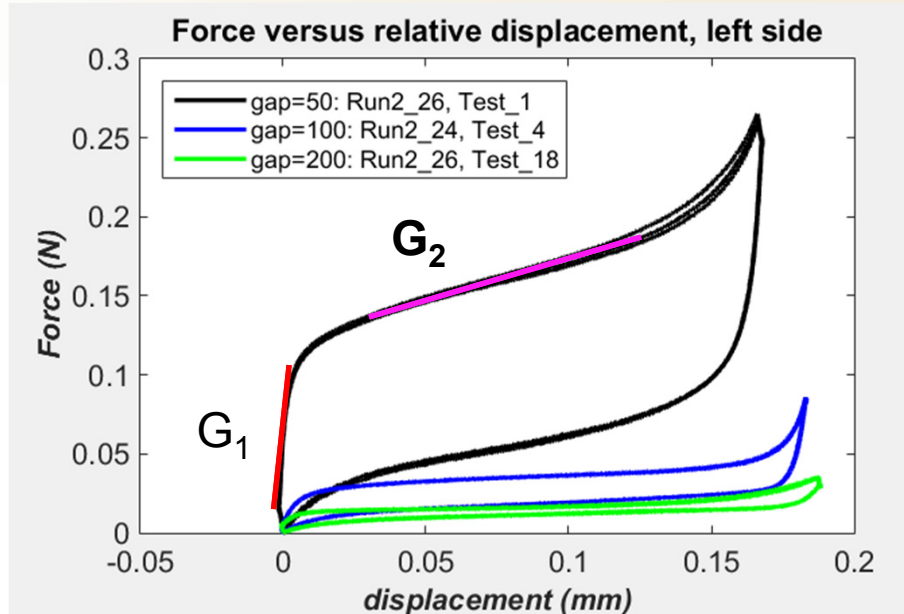
Gain test results – gap dependant; Modulus G_1 and G_2



Shear modulus G of GaIn interface

gap μm	G_{2_av} Pa	G_{2_min} Pa	G_1 Pa
50	43.7	20.0	1421
100	16.2	8.6	534
150	10.3	3.4	232
200	8.2	4.7	285
250	8.3	4.8	94

Gain test results – gap dependant; Modulus G_1 and G_2

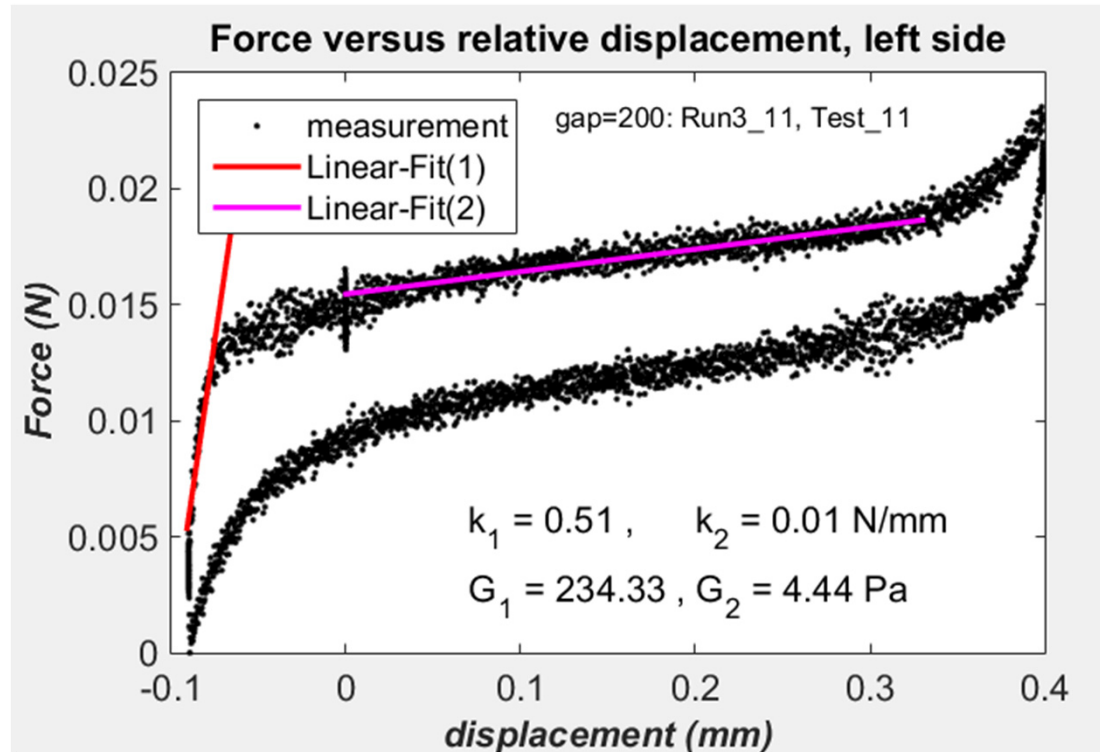


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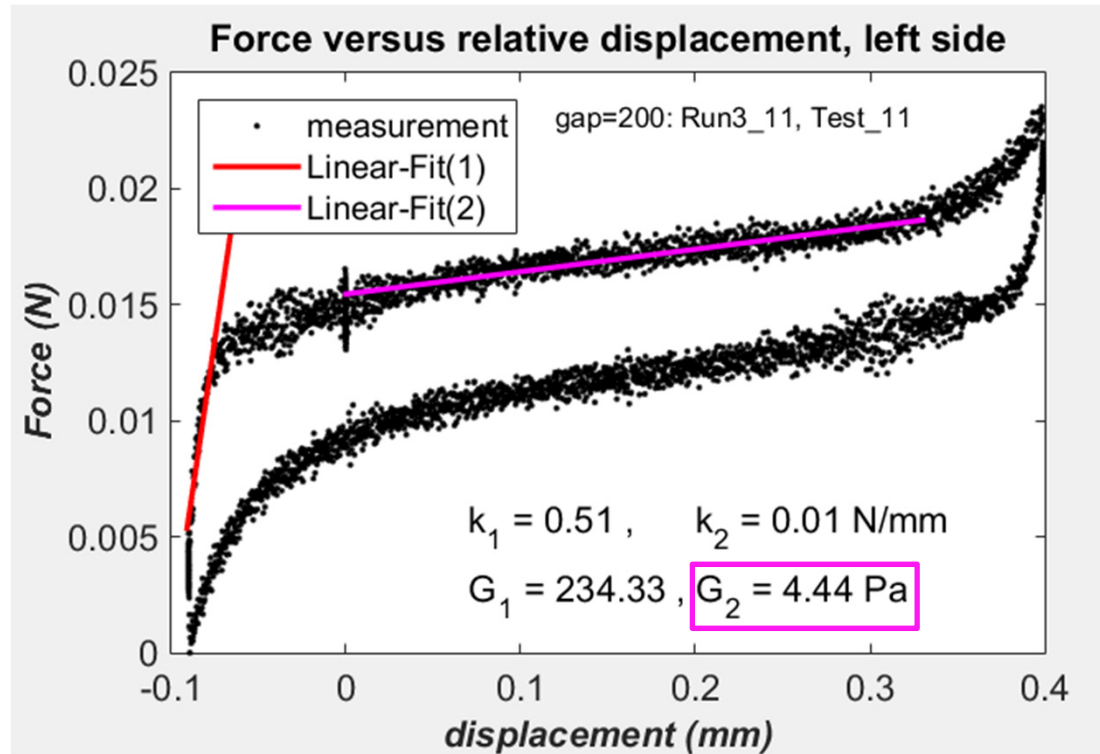
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$G_2 = 9 \text{ Pa}$

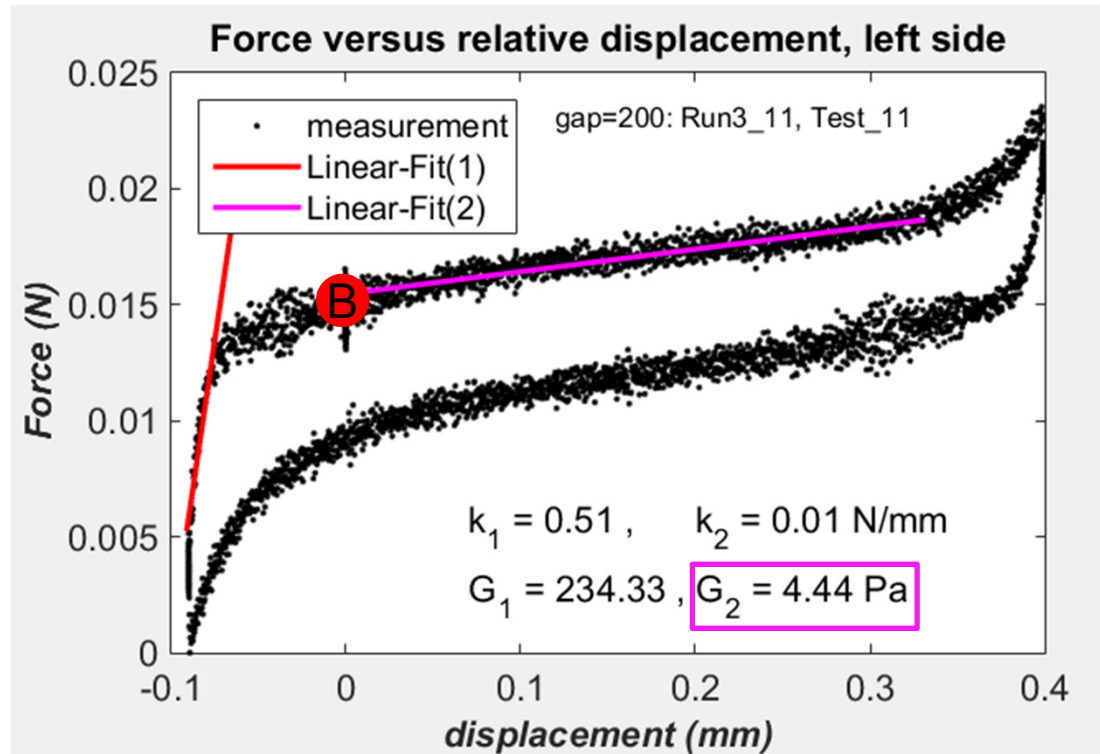
Bending Operation Optimization with Galn Interface



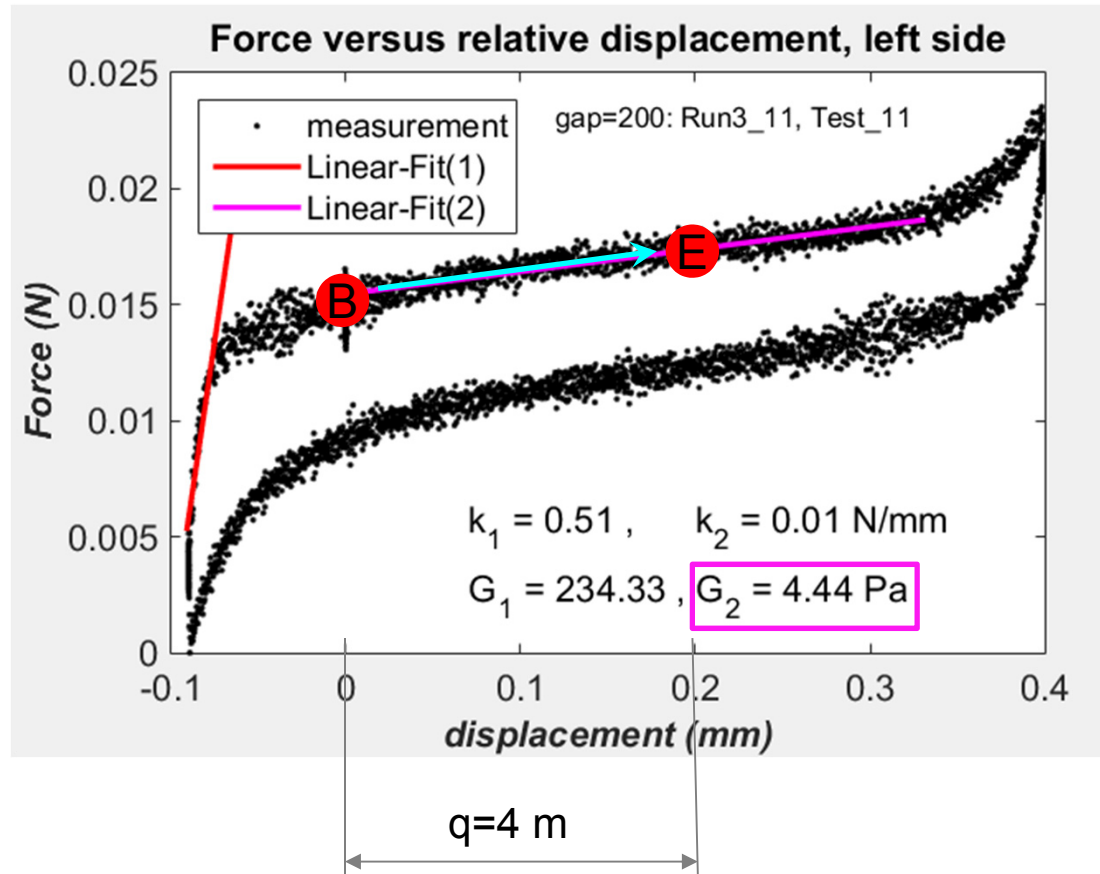
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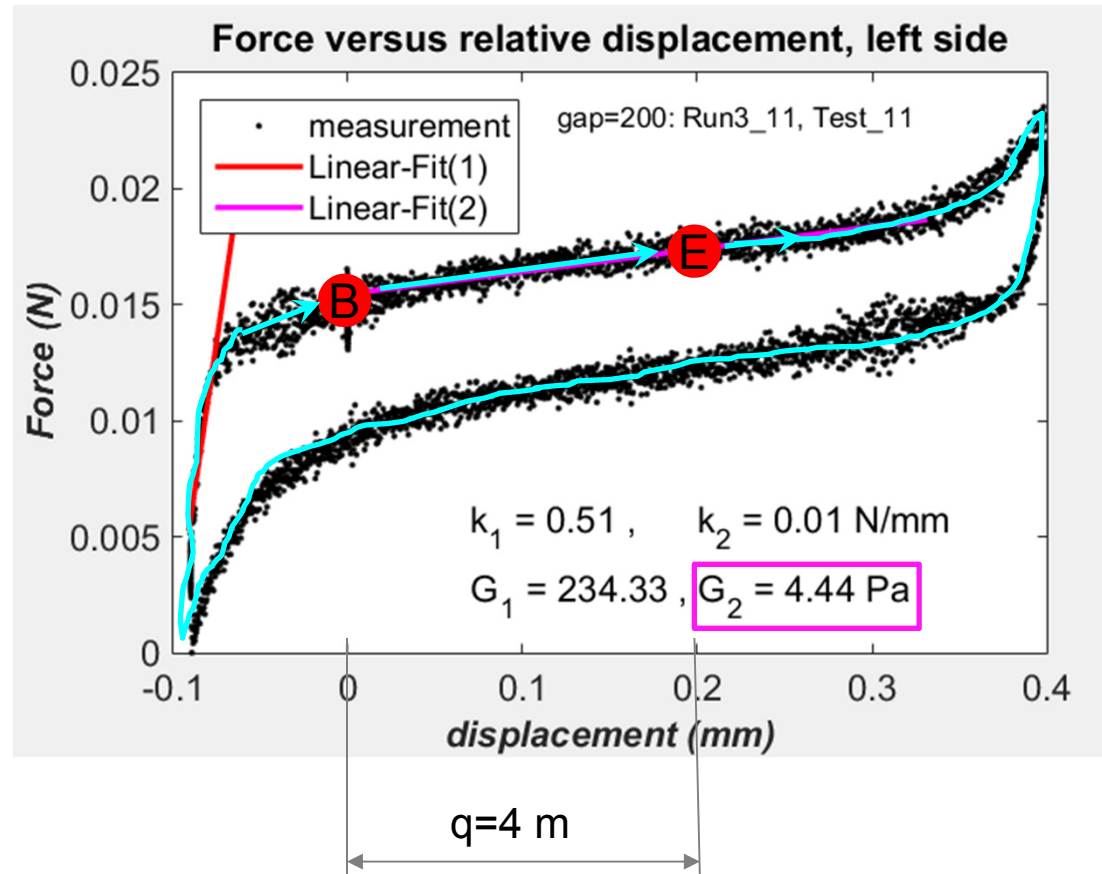
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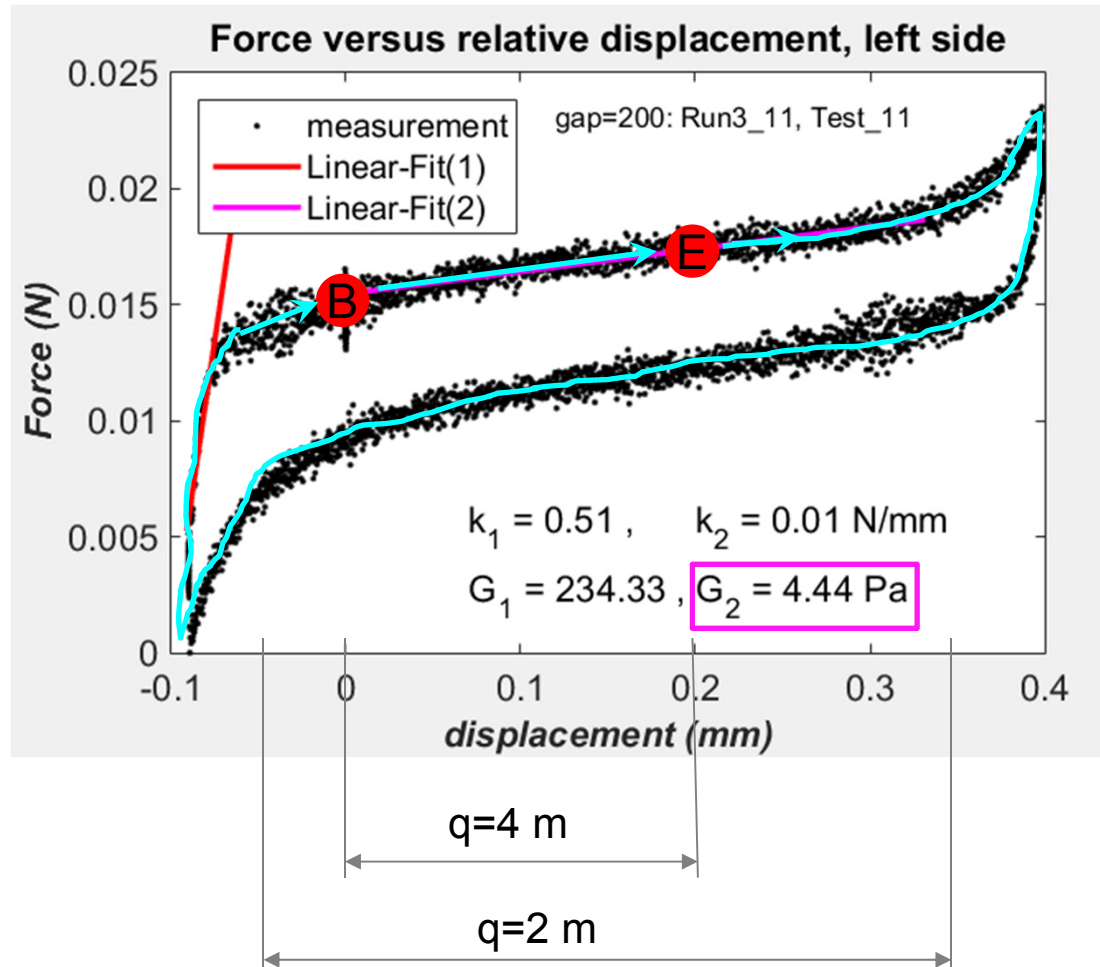
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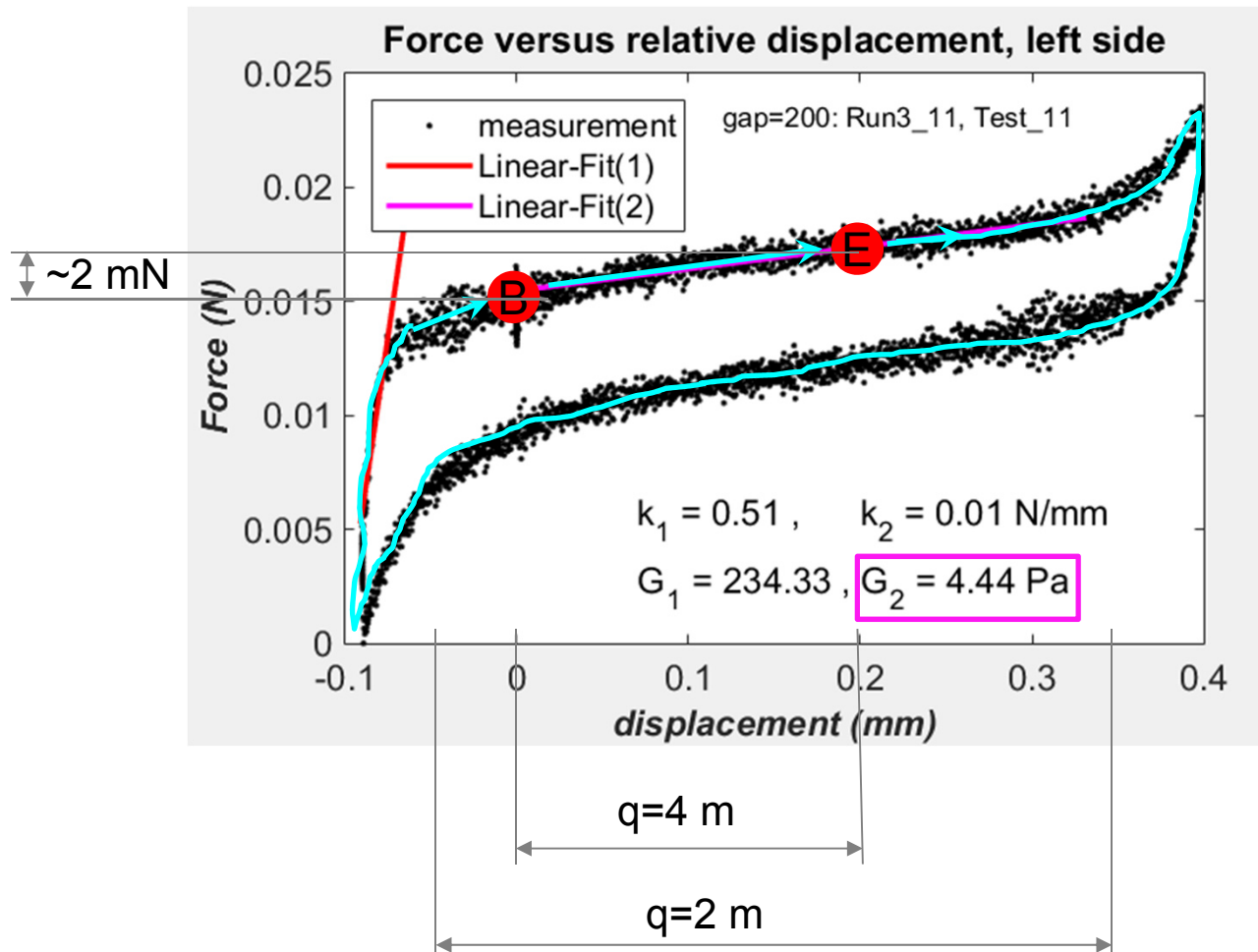
Bending Operation Optimization with Galn Interface



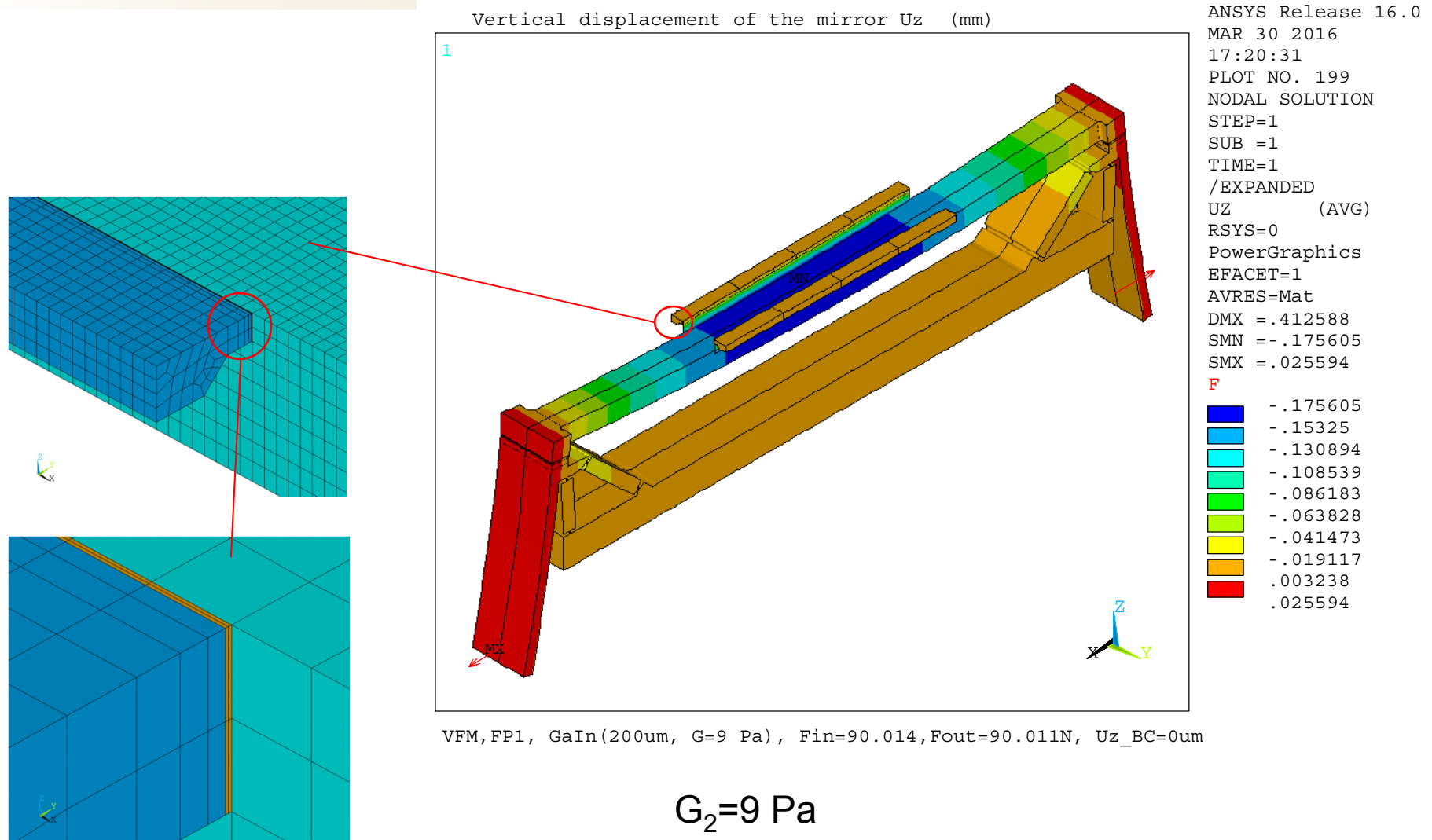
Bending Operation Optimization with Galn Interface



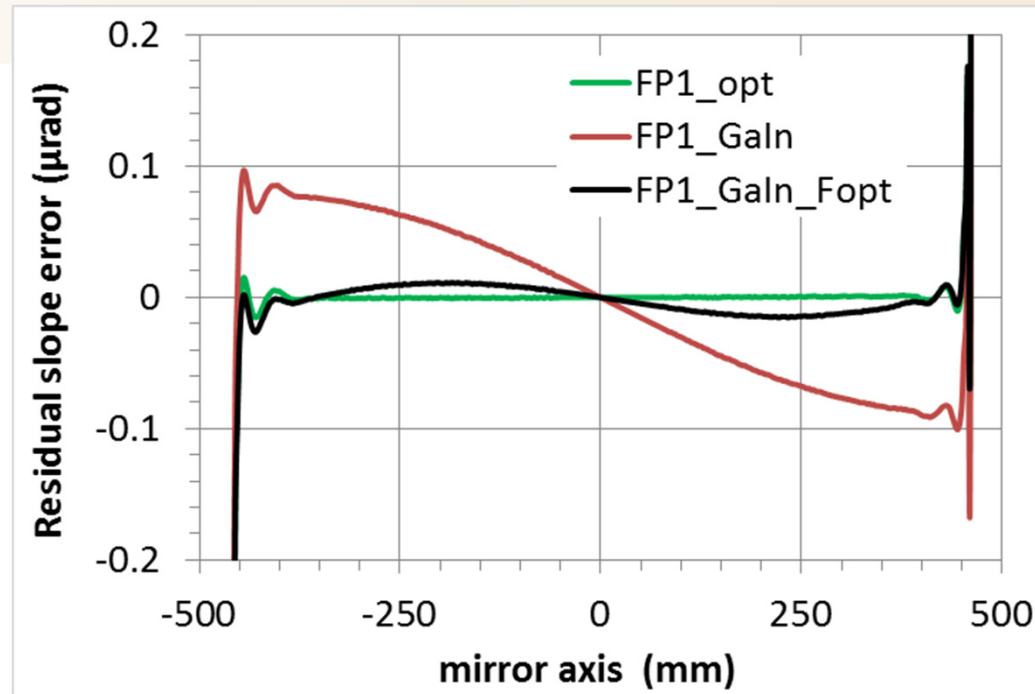
Bending Operation Optimization with Galn Interface



FEA results with cooling blocks and GaIn interface



FEA results with cooling blocks and Galn interface

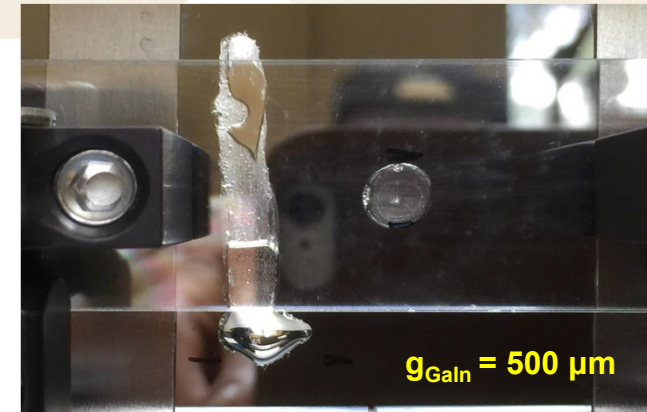
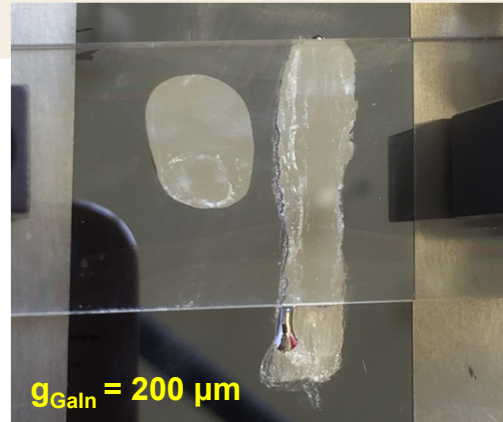
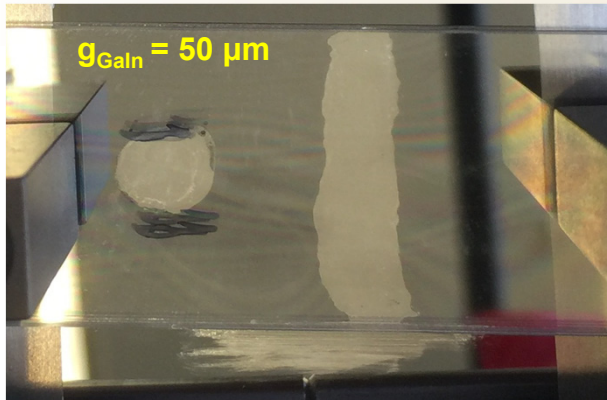


Residual Slope Error (RSE) :

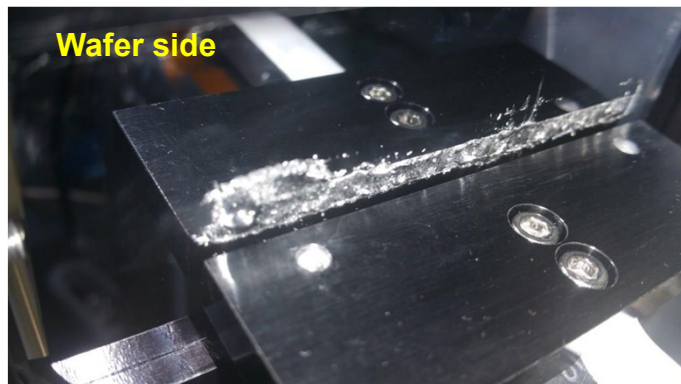
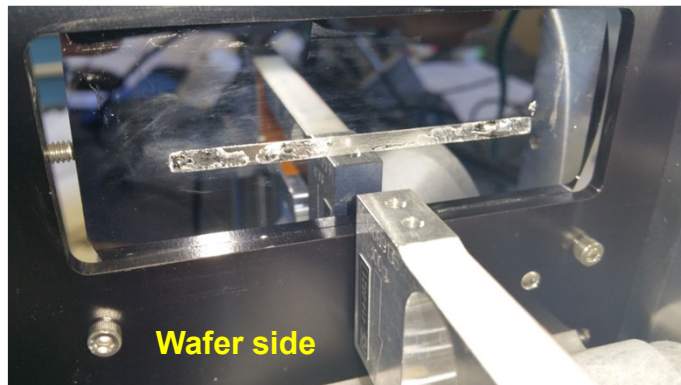
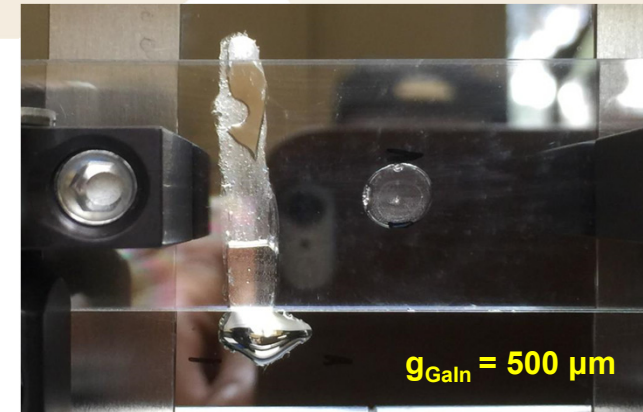
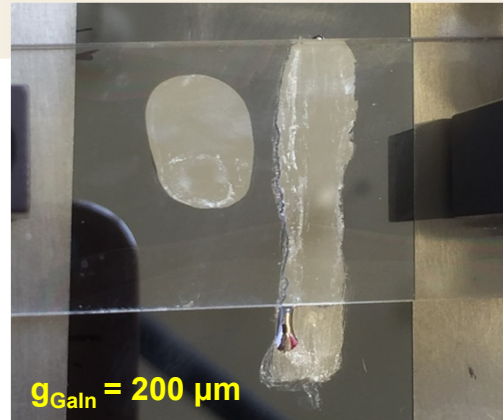
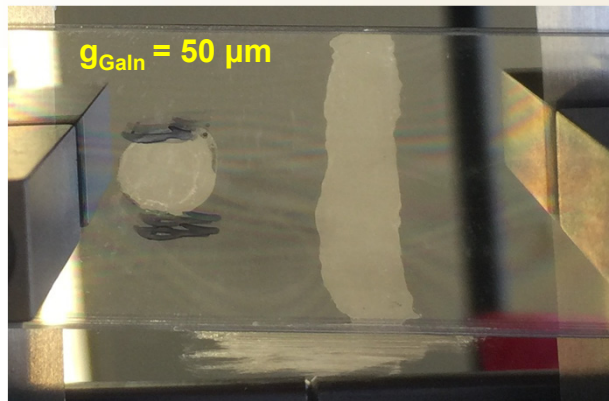
$$\Delta \text{slope} = \text{slope} - \text{slope}_{\text{ellipse}}$$

$U_{z_{BC}}$ μm	F_{Galn} mN	g_{Galn} μm	G_2 Pa	q m	F_{inp} N	F_{outp} N	RMS_{dUz} nm	RMS_{dslp} $nrad$
		-1	-1	4.7	90	90	0.4	1
0	-13.7	200	9	4.7	90	90	10.0	57
0	-13.7	200	9	4.7	90.014	90.011	1.8	9

Galn gaps



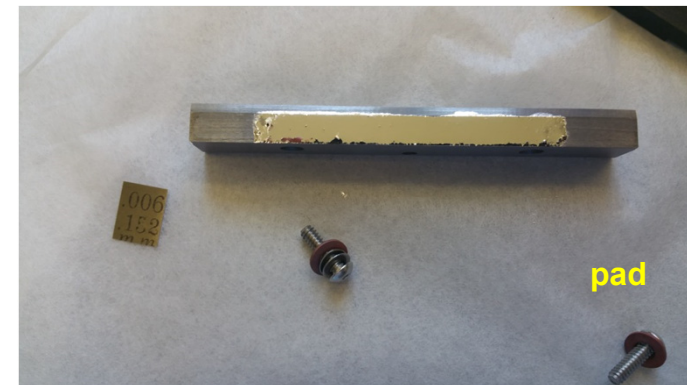
GaIn gaps



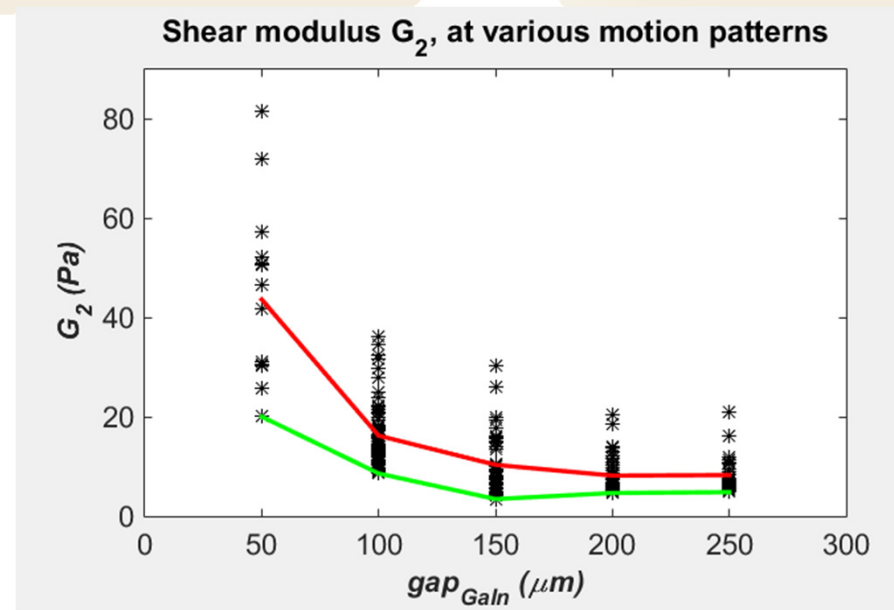
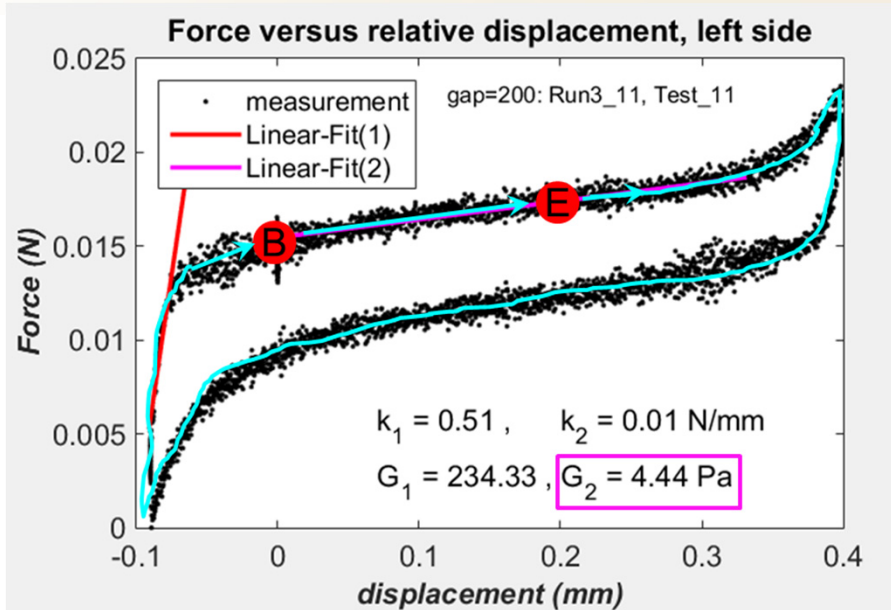
Before assembling
for test at $g_{\text{GaIn}} =$
 $152 \mu\text{m}$

After test at
 $g_{\text{GaIn}} = 152 \mu\text{m}$

in as thermal interface



Summary of Galn test results



- The mechanical constraint effects of Galn contact interface can be minimized to be negligible for LCLS-II SXR KB mirrors by
- Setting the $g_{\text{Galn}} = 150 \mu\text{m}$. Shear modulus $G_2 = 9$ Pa easily achievable
 - Operating between point B and E, and following the force $\sim$ displacement curve

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- **Tom Rabedeau** SLAC/SSRL
- **Eliazar Ortiz** SLAC/LCLS-II
- **Venkat Srinivasan** SLAC/LCLS, now India
- **Peter Stefan** SLAC/LCLS